

THE MARVELS OF OUR SUN

Dr Lauren Doyle & Cam Heather

What are we going to cover today?

- **SOLAR I** - *Introduction, getting to know the Sun, and computing some simple Python calculations*
- **SOLAR II** - *Accessing and manipulating solar data using the SunPy package, creating maps of the solar surface*
- **SOLAR III** - *Looking into solar flares through time series data*
- **SOLAR IV** - *Investigating what stellar flares look like on other stars*

A little bit about us...



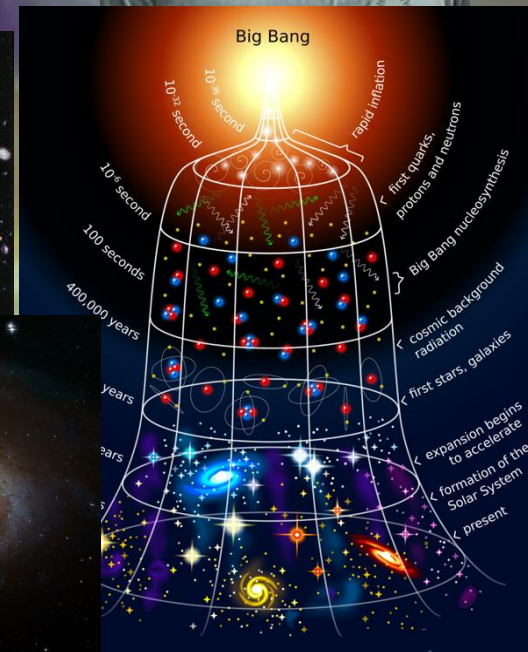
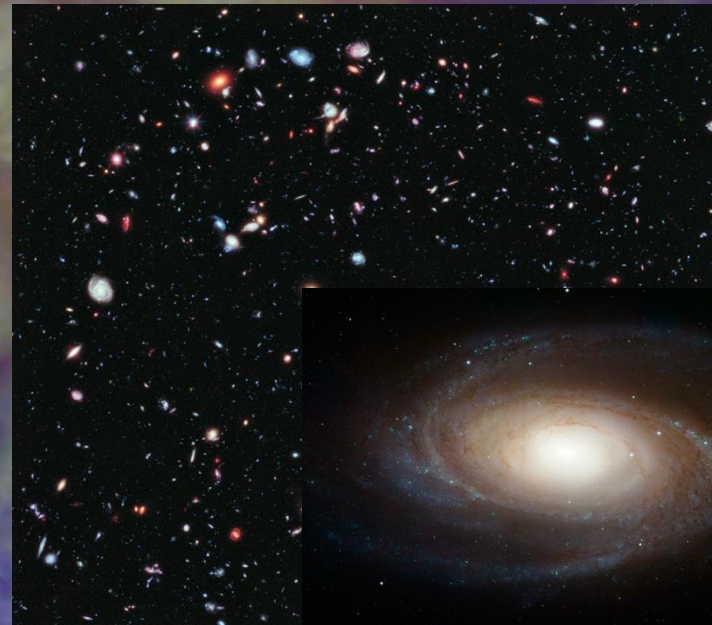
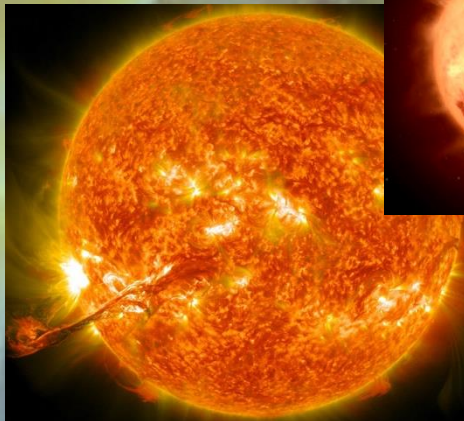
Dr Lauren Doyle
(she/her)



From Glasgow,
Scotland

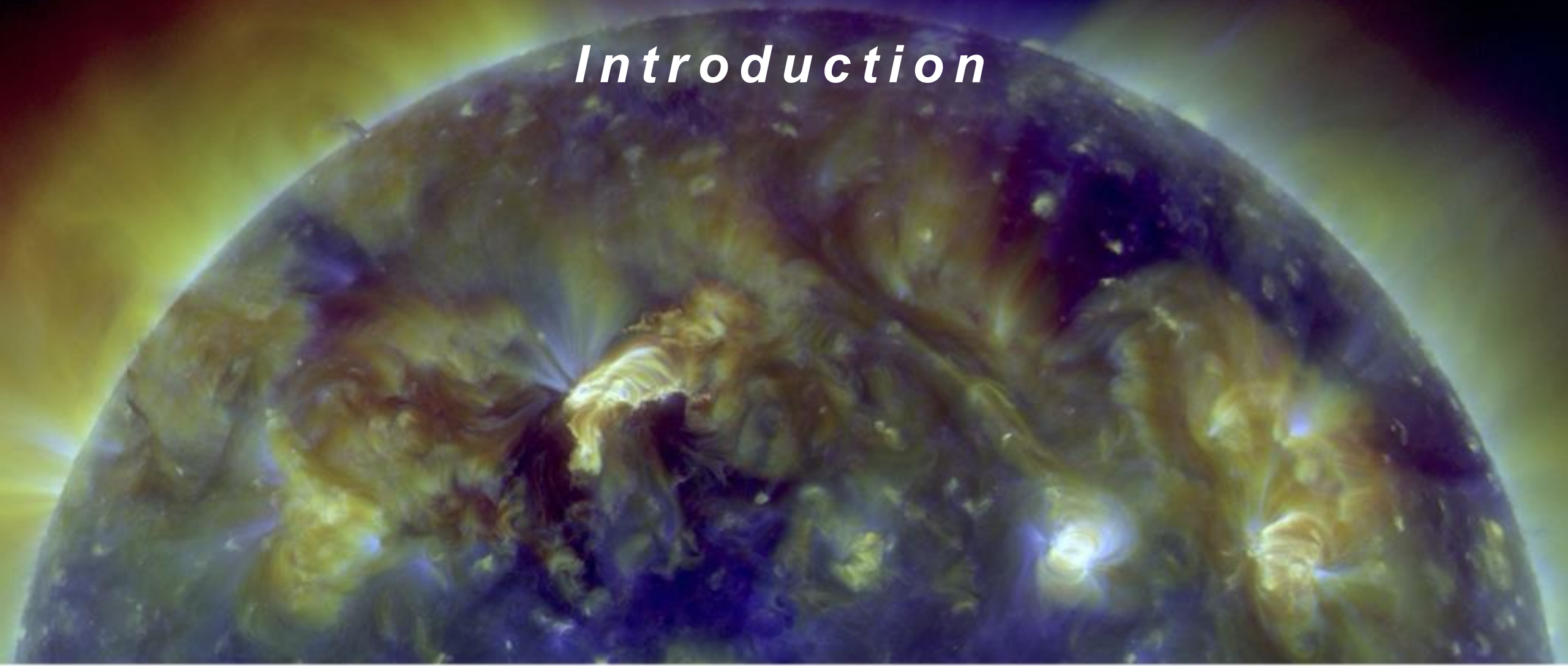
Cam Heather
(she/her)

4th year PhD student



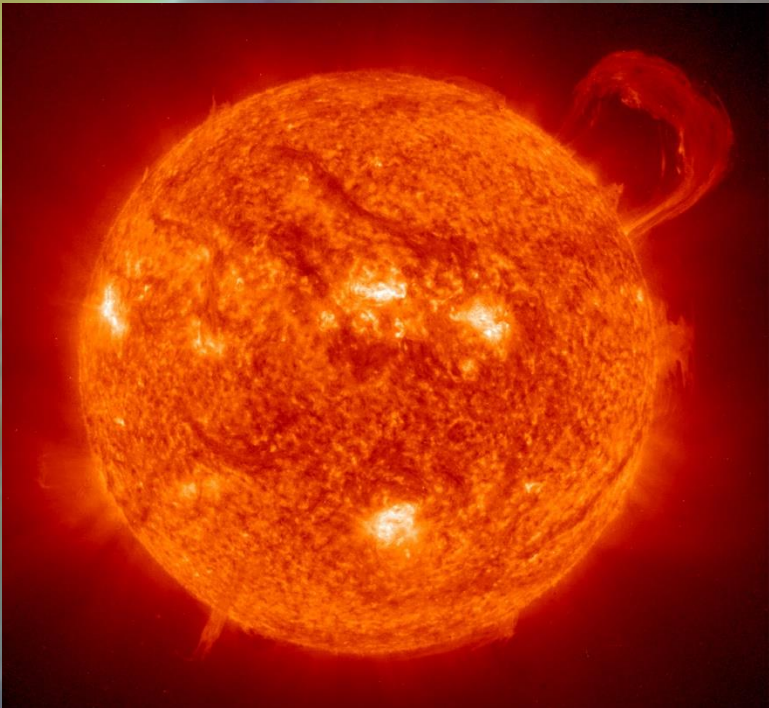
SOLAR I

Introduction



Quick Sun Facts:

- The Sun is our nearest star!
- It is made up of Plasma.
- It contains 99.86% of the mass of the entire solar system.
- The Sun is an average-sized star.



Hydrogen: 71%

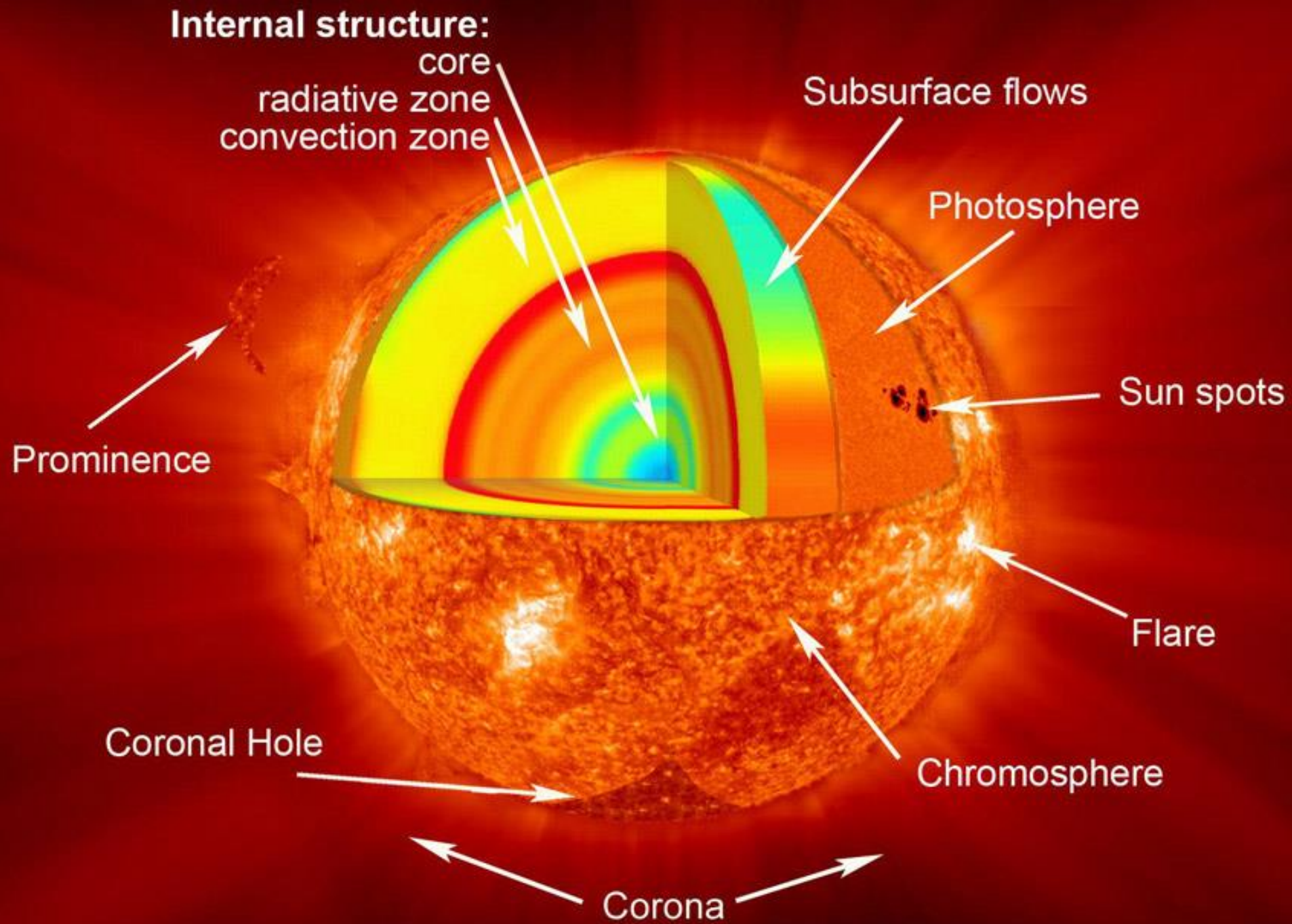
**What is
the Sun
made
of?**

**Helium:
27%**

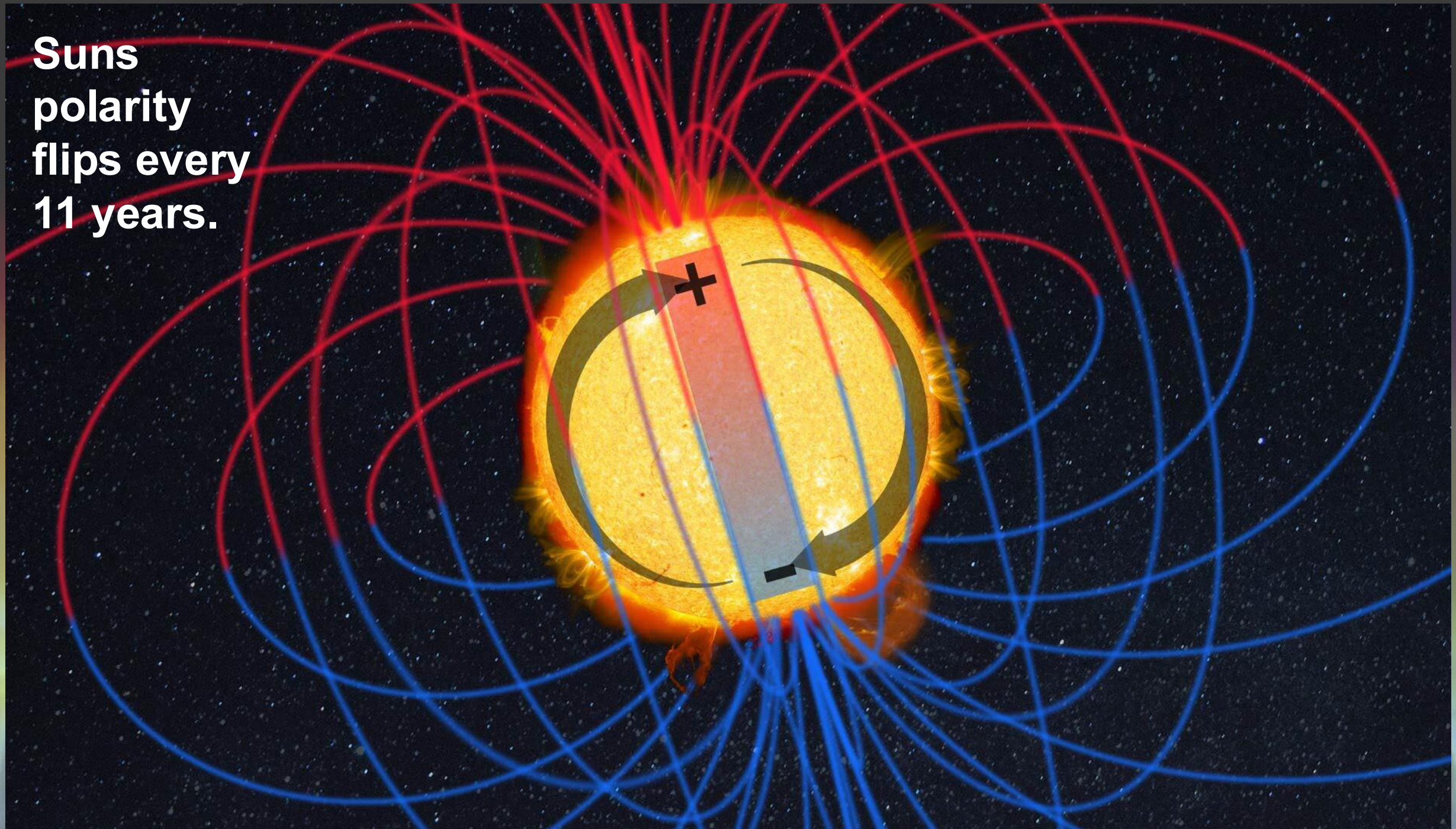
**Everything
else: 2%**

Solar Temperatures

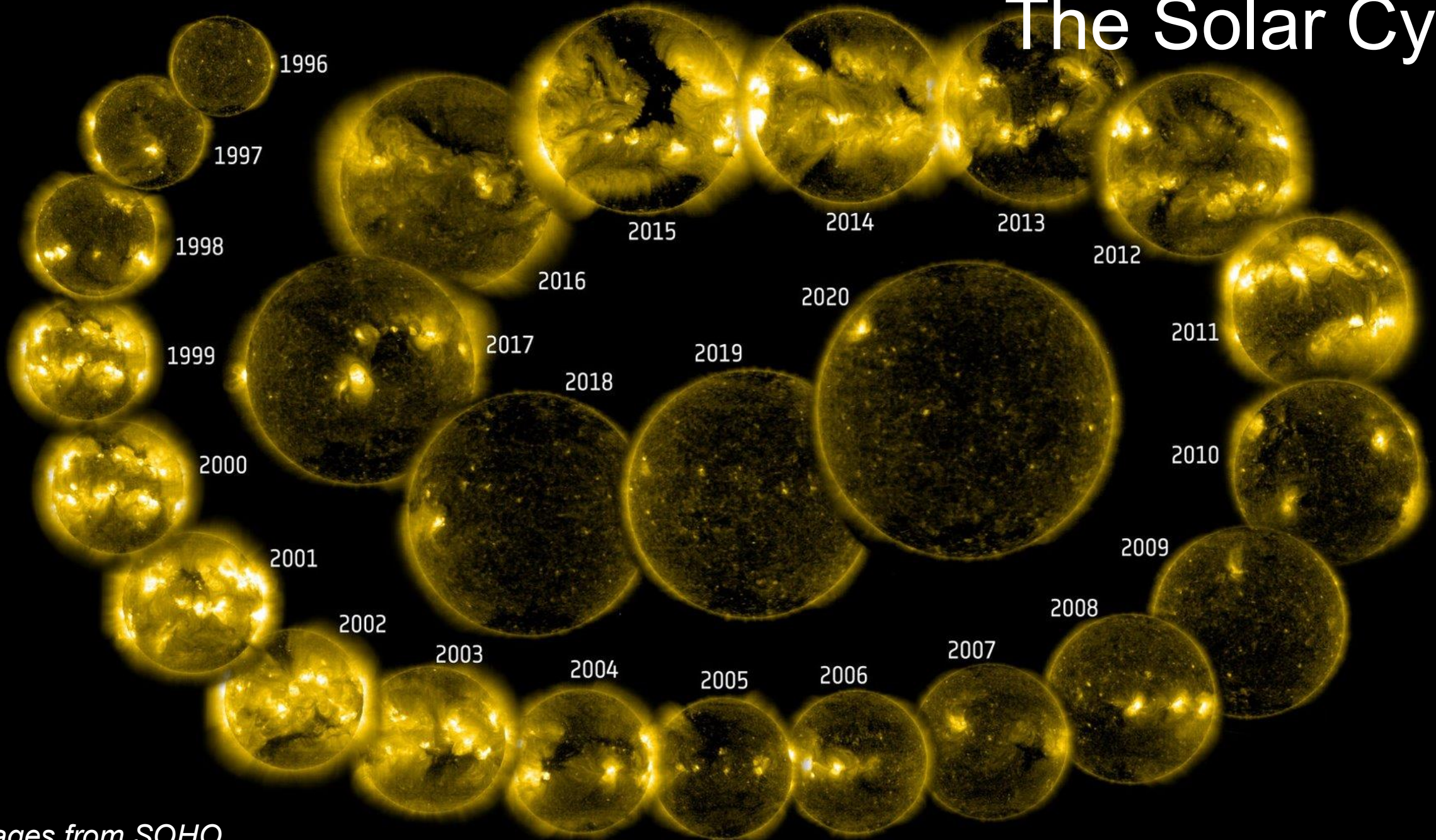
- Core temperature: 15,000,000 K
 - Surface temperature: 6,000 K
 - Corona temperature: 2,000,000 K
- (1 K = -272.15 °C)



**Sun's
polarity
flips every
11 years.**

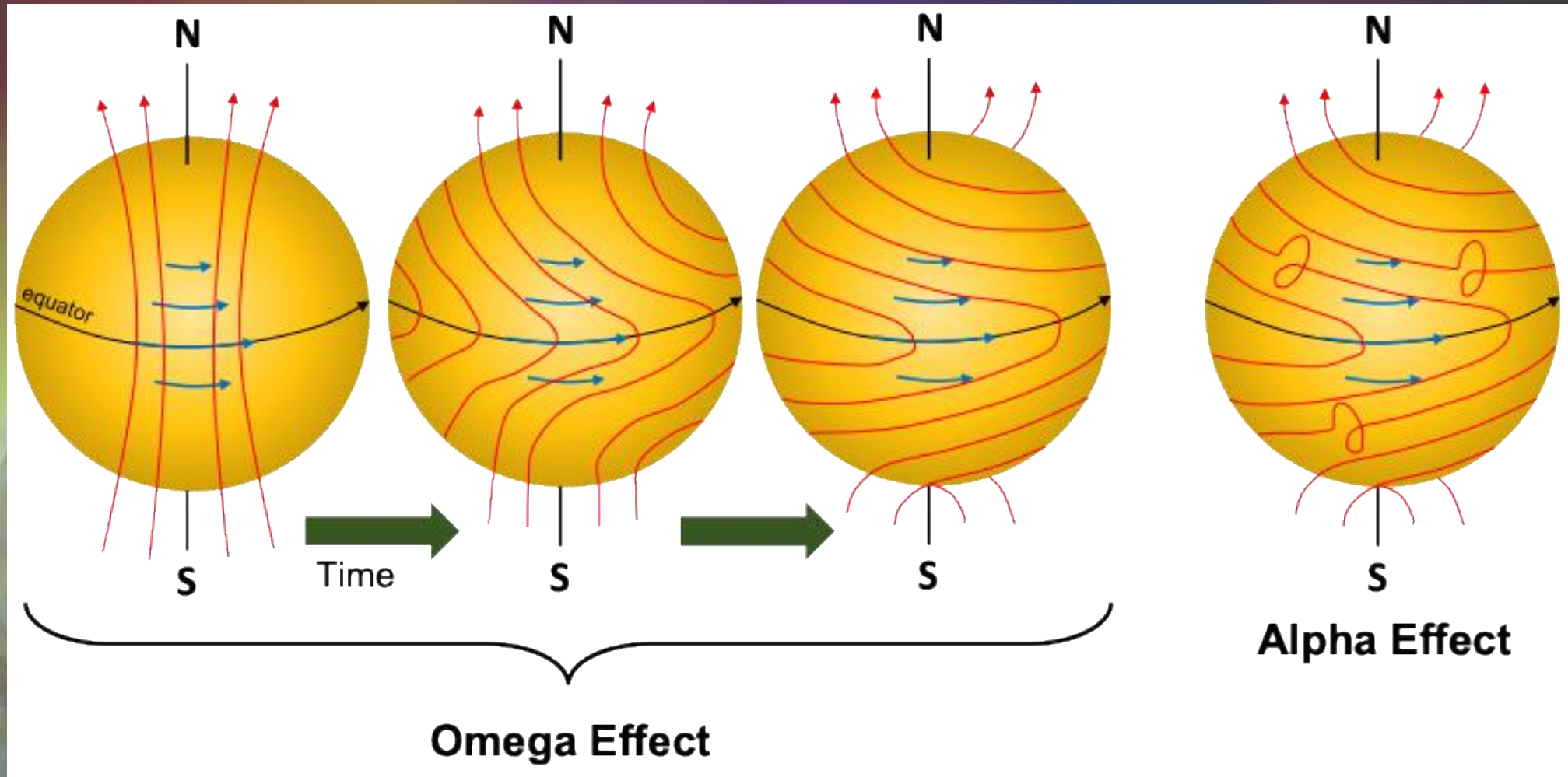


The Solar Cycle



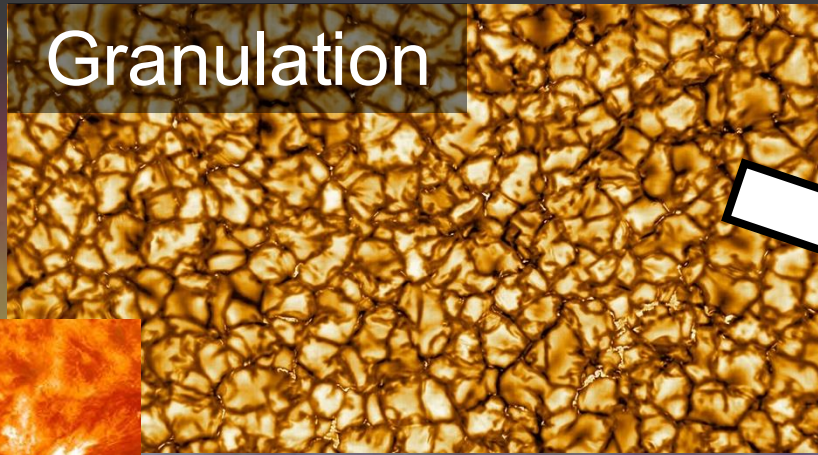
Images from SOHO

Solar Dynamo

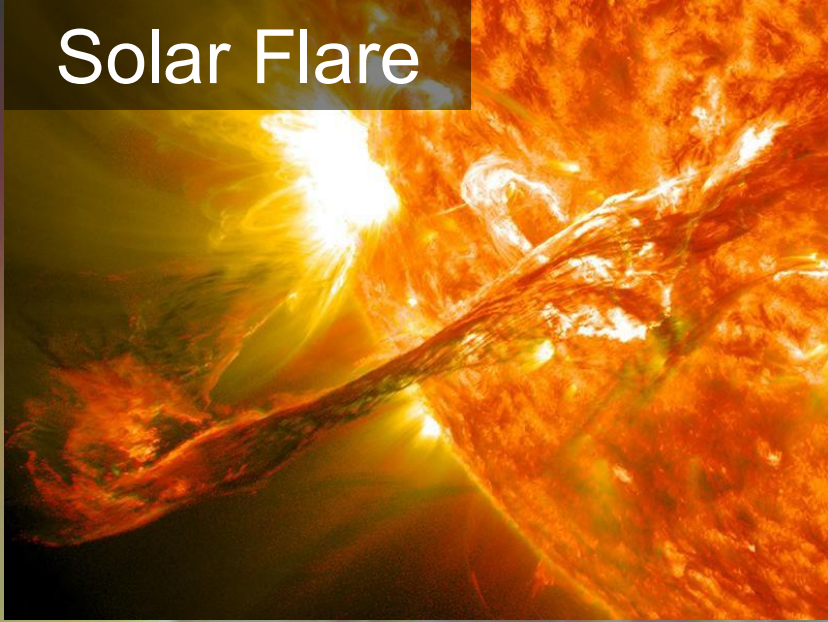


Solar Phenomena

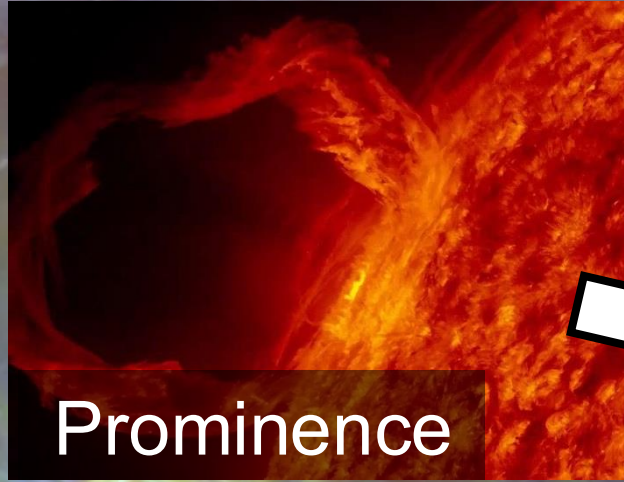
Granulation



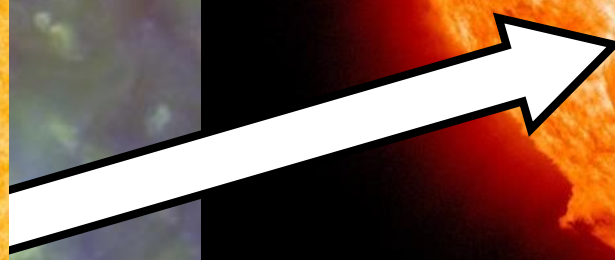
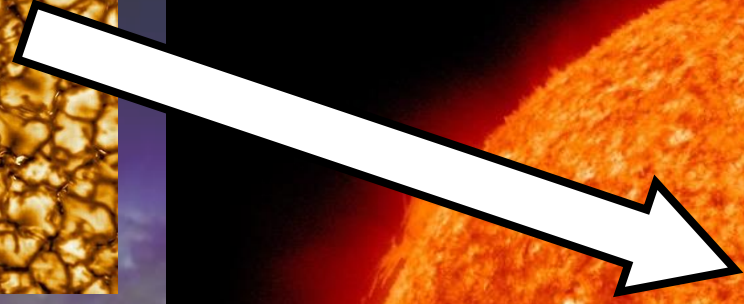
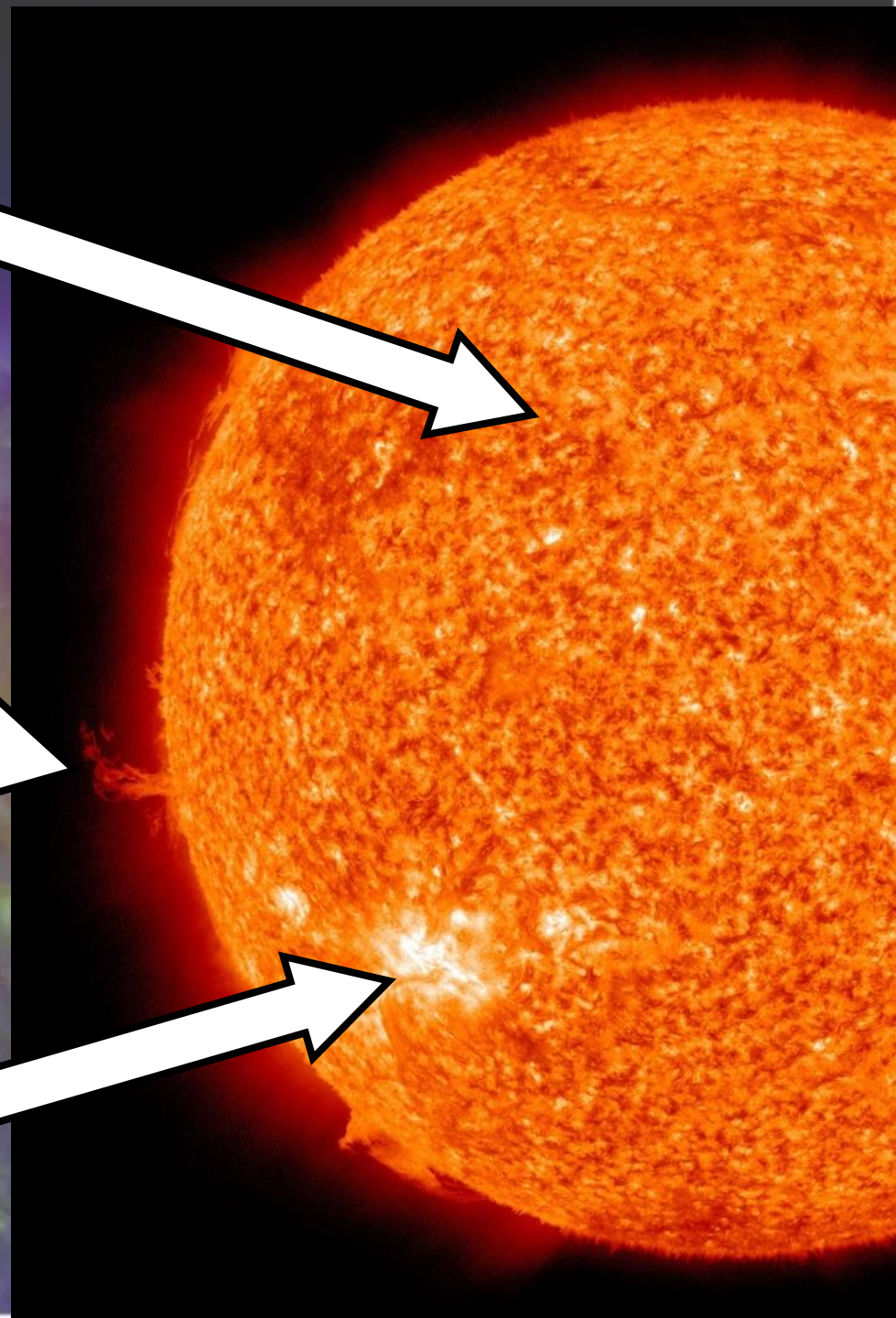
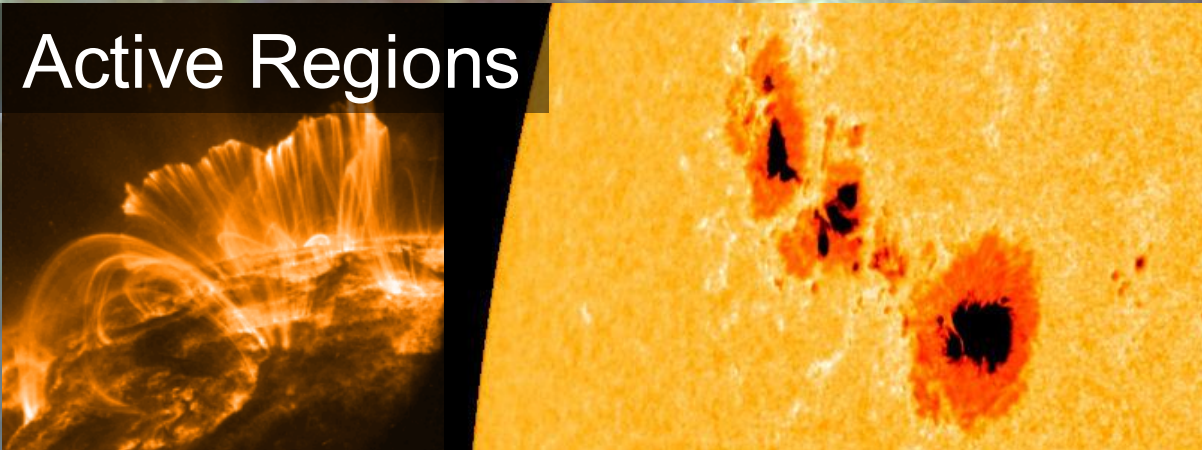
Solar Flare



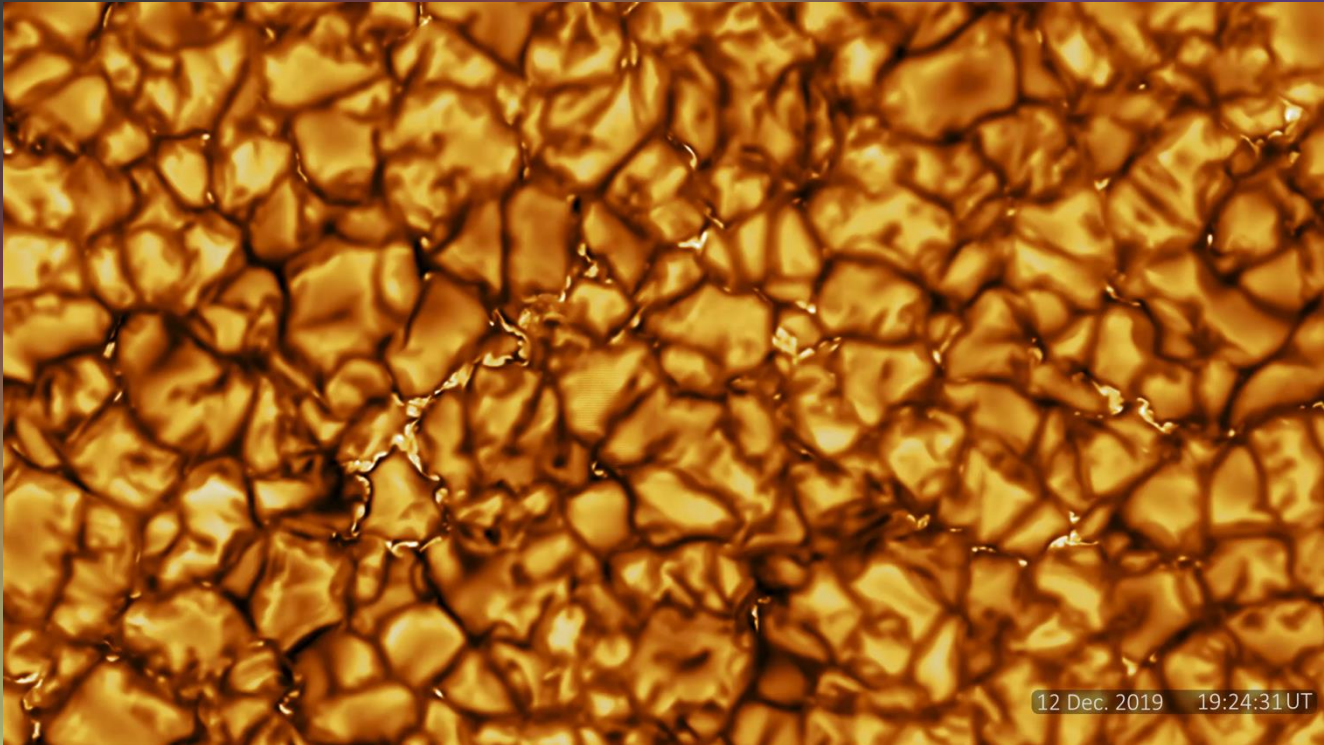
Prominence



Active Regions

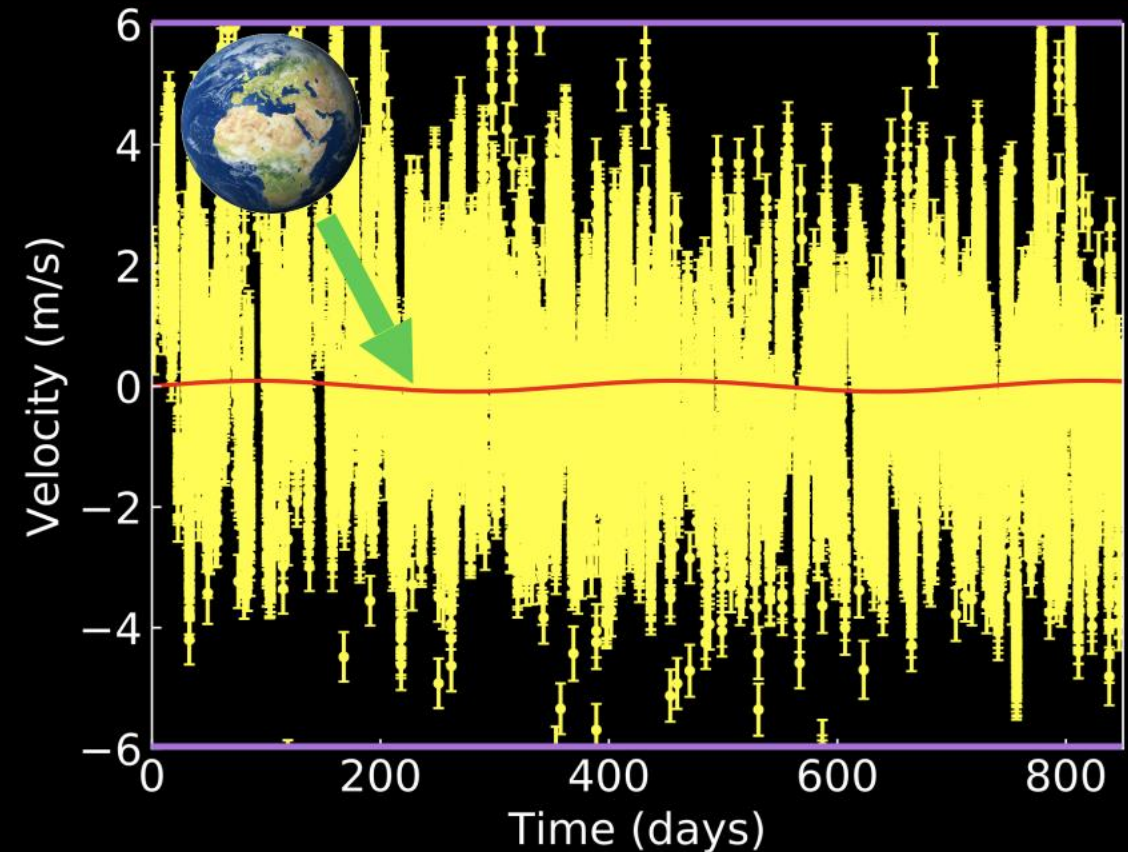


Granulation



Surface is covered in
bubbles of plasma!

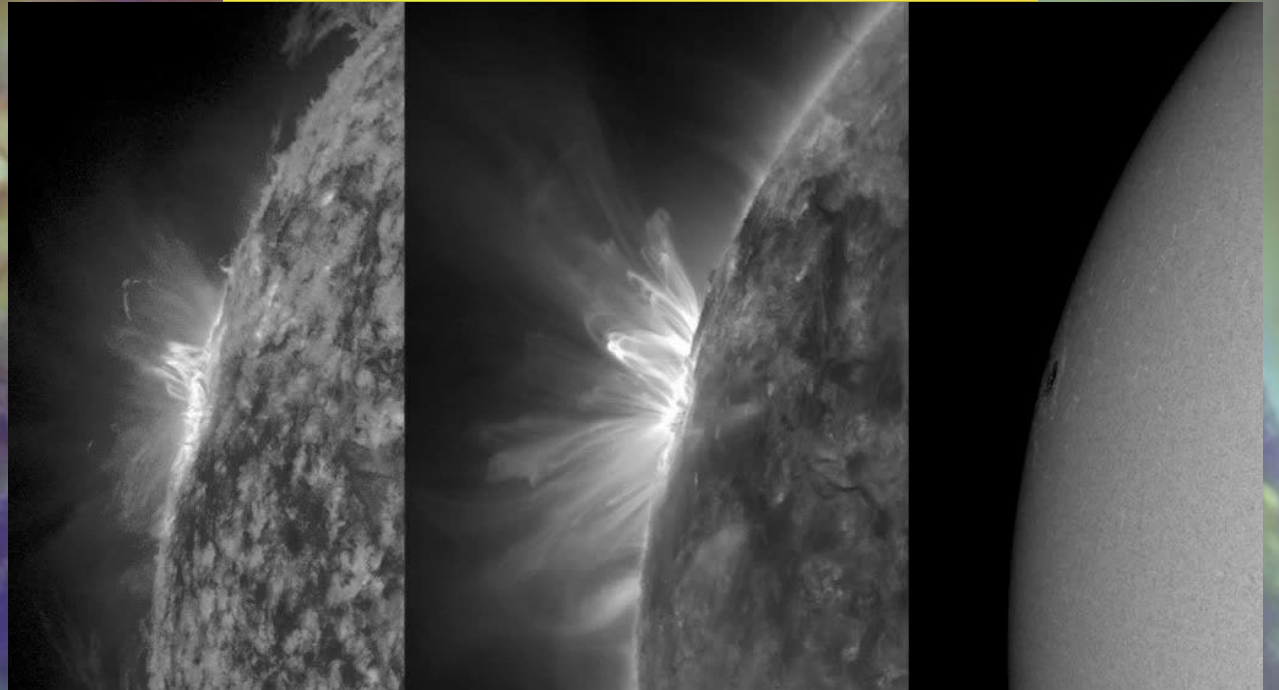
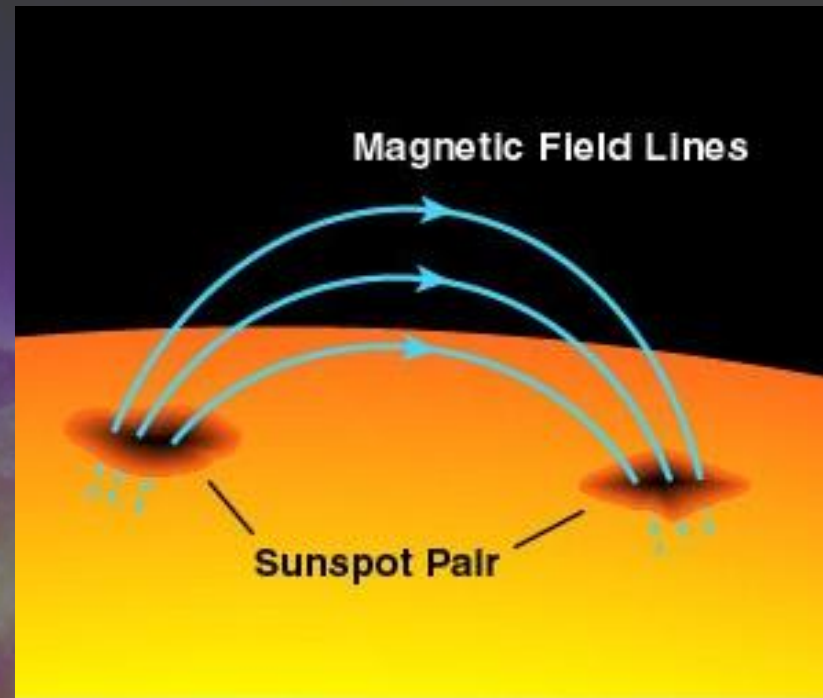
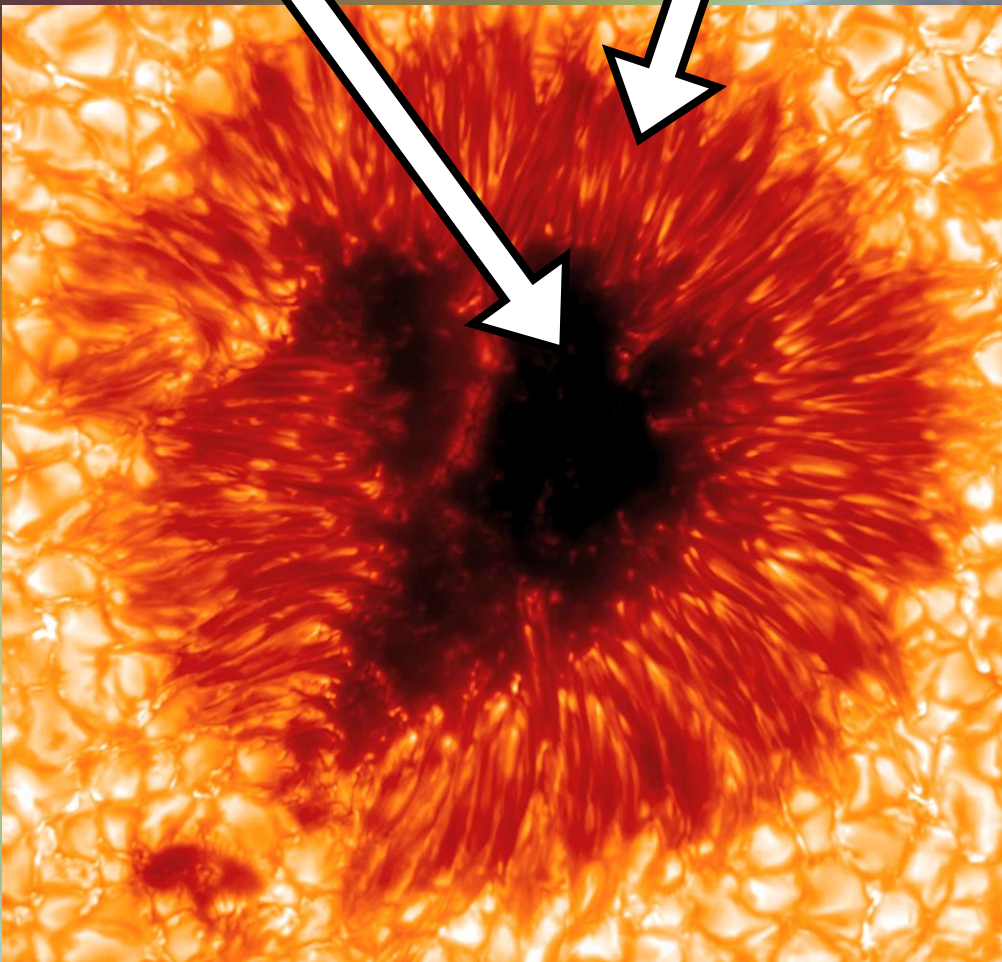
Creates stellar noise which
makes it difficult to detect
small earth like planets!



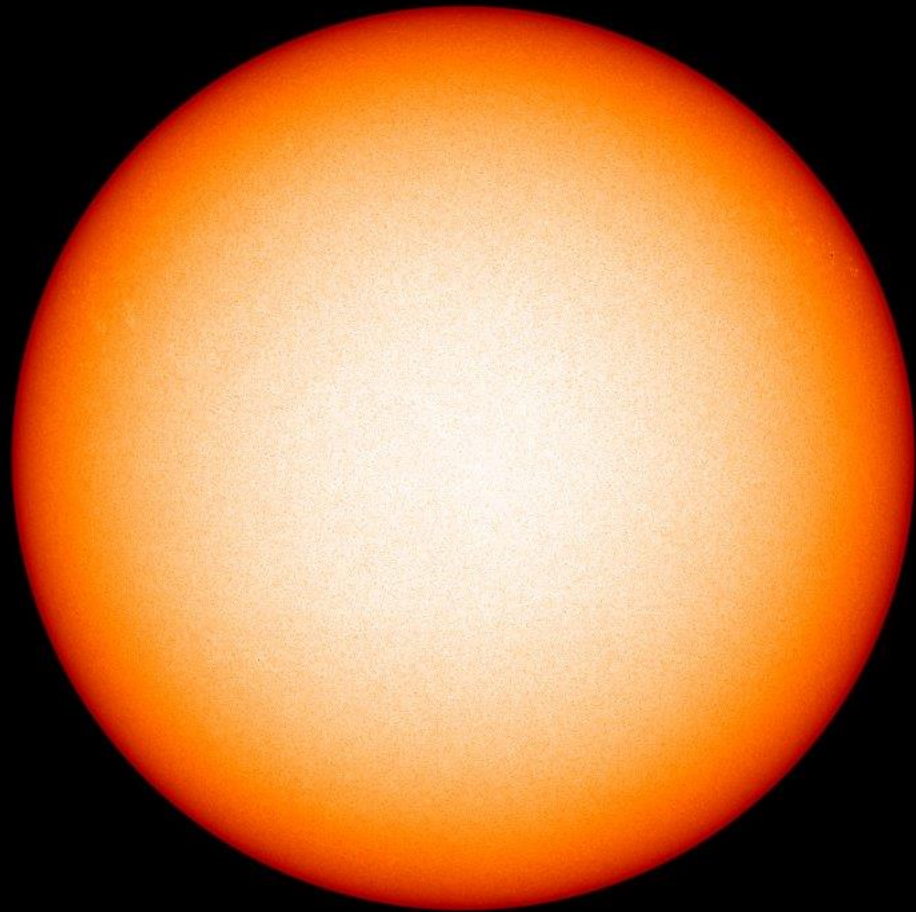
Sunspots

Penumbra

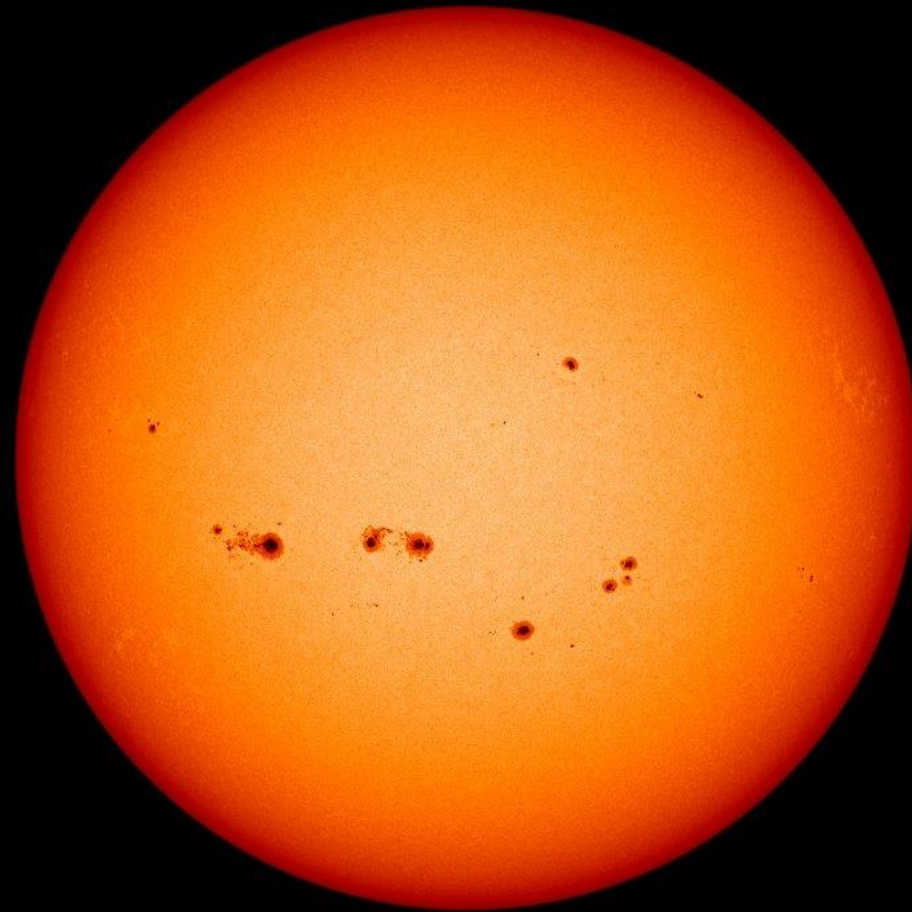
Umbra



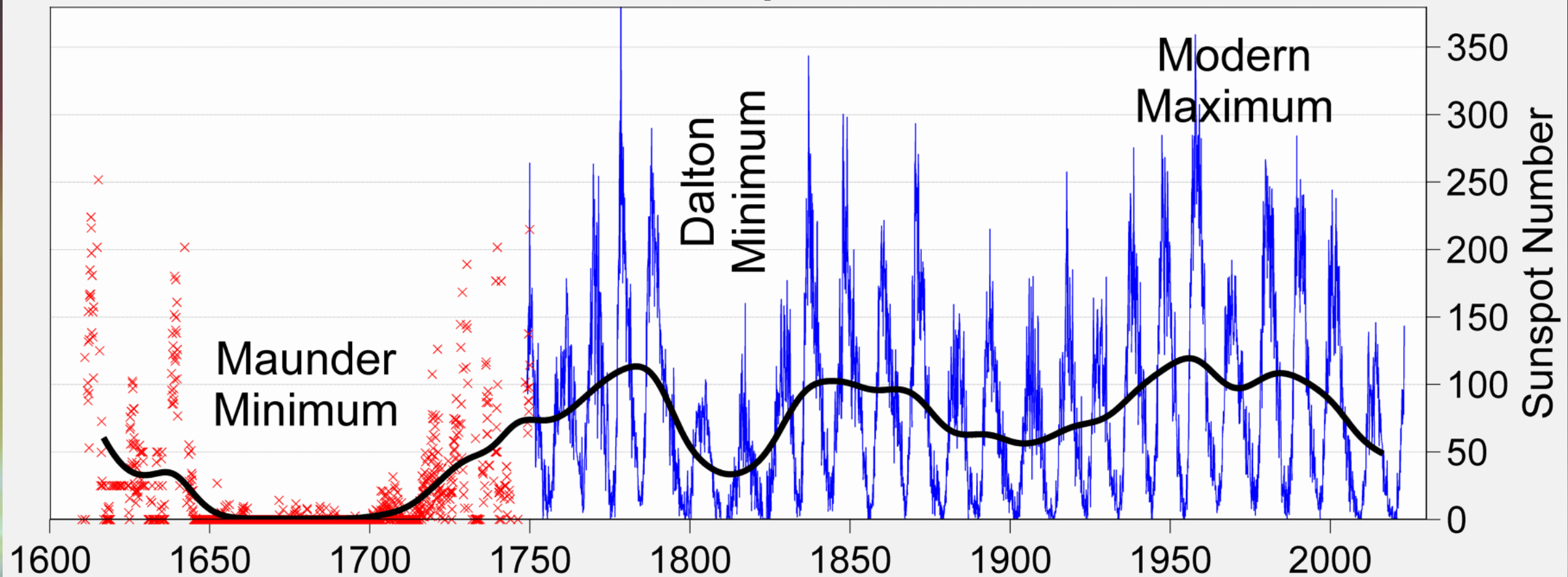
SOLAR MINIMUM



SOLAR MAXIMUM



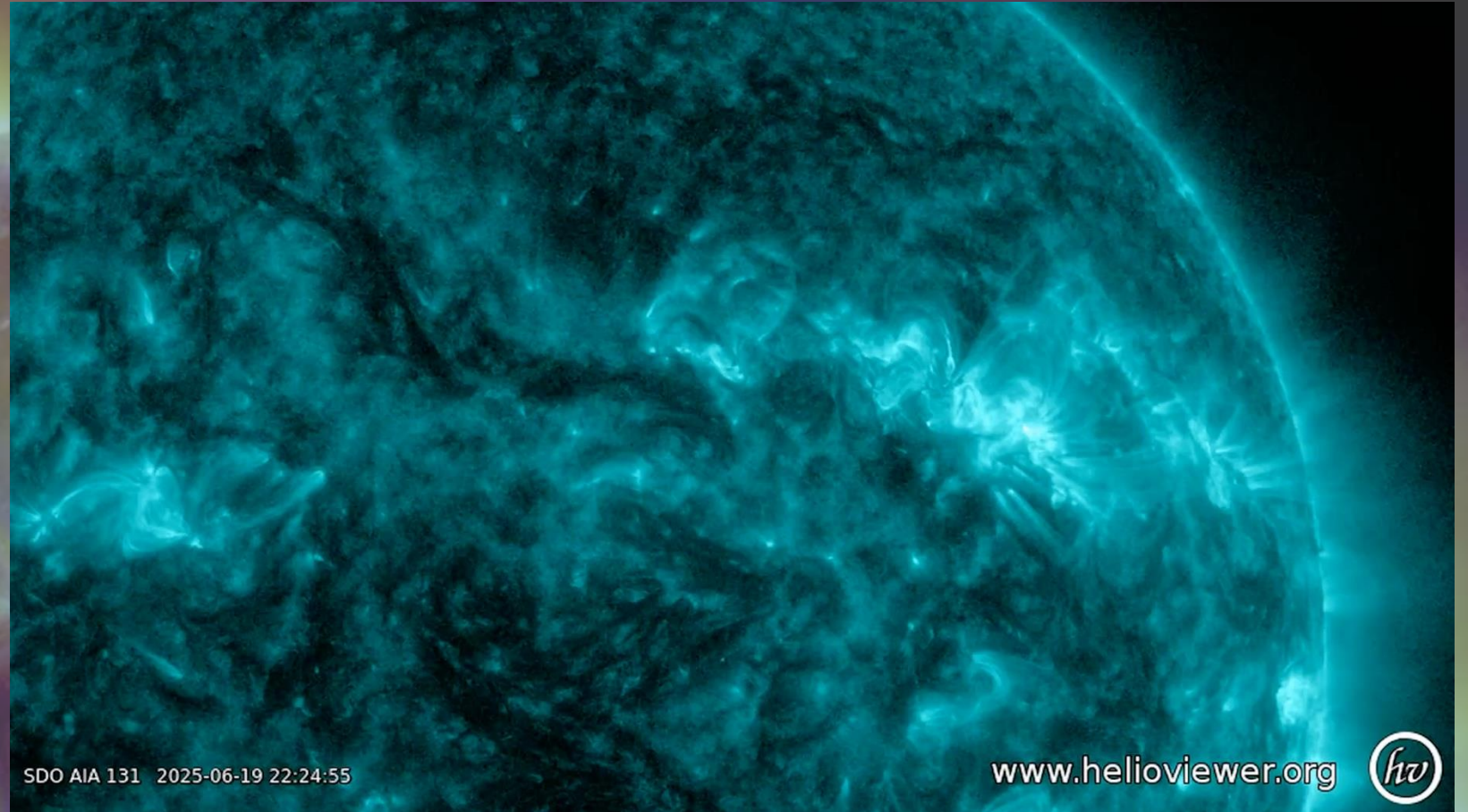
400 Years of Sunspot Observations



Solar Flares

A solar flare is an intense, localised brightening caused by the emission of electromagnetic radiation in the Sun's atmosphere.

- Spans the entire electromagnetic spectrum
- Often accompanied by a CME.



SPACE WEATHER IMPACTS

AURORA
(NORTHERN LIGHTS)



GPS



SATELLITES



COMMUNICATIONS



HUMAN SPACE
EXPLORATION



AVIATION



ELECTRIC POWER



WEATHER.GOV/SPACE

Top Tips & Reminders:

Ask for help!

If something isn't working or doesn't make sense, please let us know — don't sit stuck.

Read all the instructions carefully

The notes contain important hints and explanations that will help you complete the exercises.

Run each cell in order

Skipping cells can lead to errors later on.

Read error messages

They often tell you exactly what's wrong — take a moment to look before asking for help.

Try things out!

Don't be afraid to experiment — that's how you learn.

Save your work regularly

Especially if you're working online (e.g. Colab).

Check your file paths

Many errors come from Python not being able to find your files.

Focus on understanding, not just getting the answer

The goal is to learn how and why things work.

Work together, but write your own code

Discuss ideas but make sure you understand what you submit.

Take your time

It's normal for things to take a bit of thinking — that's part of coding!

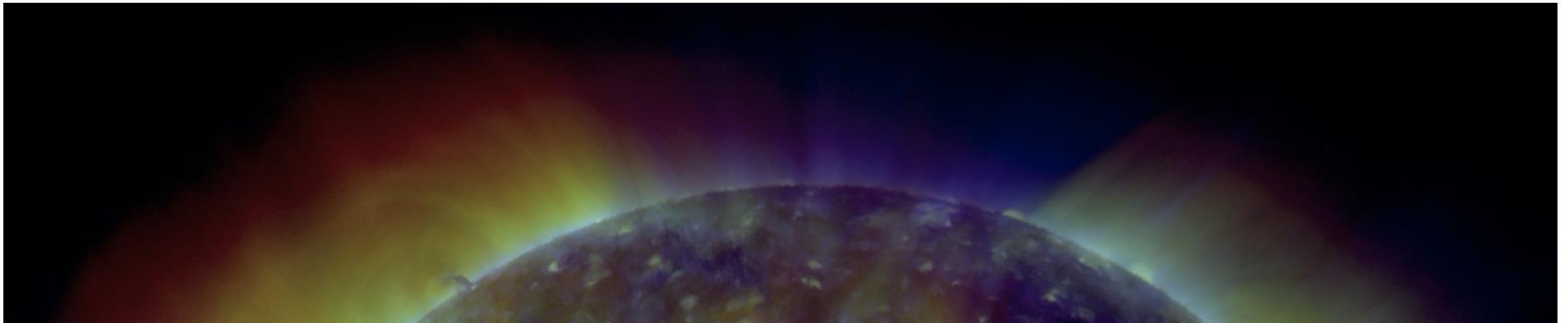
SOLAR I - Introduction

Welcome to the ThaiPASS 2025 Solar and Stellar Physics Workshop!

Today, we will cover four lessons delving into the world of Solar Physics and Stellar Physics. We will access and manipulate real solar data from the Solar Dynamics Observatory to study solar phenomena in multiple wavelengths. We will then explore how these phenomena can be studied on other stars using data from the Transiting Exoplanet Survey Satellite. The breakdown of the day is as follows:

- **SOLAR I** - Introduction, getting to know the Sun, and computing some simple Python calculations
- **SOLAR II** - Accessing and manipulating solar data using the SunPy package, creating maps of the solar surface
- **SOLAR III** - Looking into solar flares both through time series data
- **SOLAR IV** - Investigating what stellar flares look like on other stars

```
[10]: display(Image(filename="aia_header.png", embed=True))
```



GOAL : In this session we will look at some simple calculations to determine physical properties of the Sun and explore the concept of time in astronomy.

AIM : To practice using Python in the notebook format, compute some simple calculations and become familiar with images of the Sun.

OUTPUT : Properties of the Sun and timing conventions which will be used throughout the other sessions.

SOLAR I - Reflection

1. What was the distance between the Sun and the Earth?

149400000 km = 1.494×10^8 km = 1 AU

2. What was the radius of the Sun?

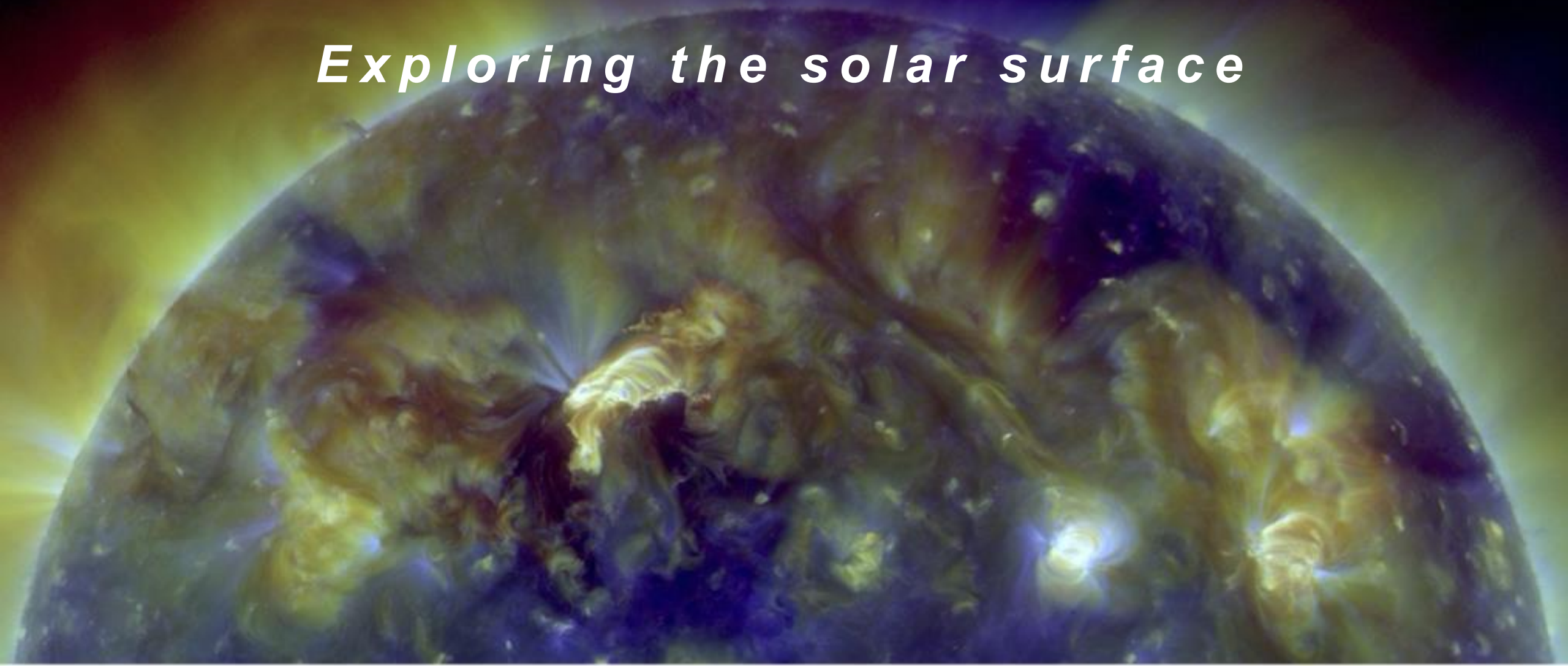
~ 699 340 km

3. How many Earths fit into the Sun?

~ 1.3 million

SOLAR II

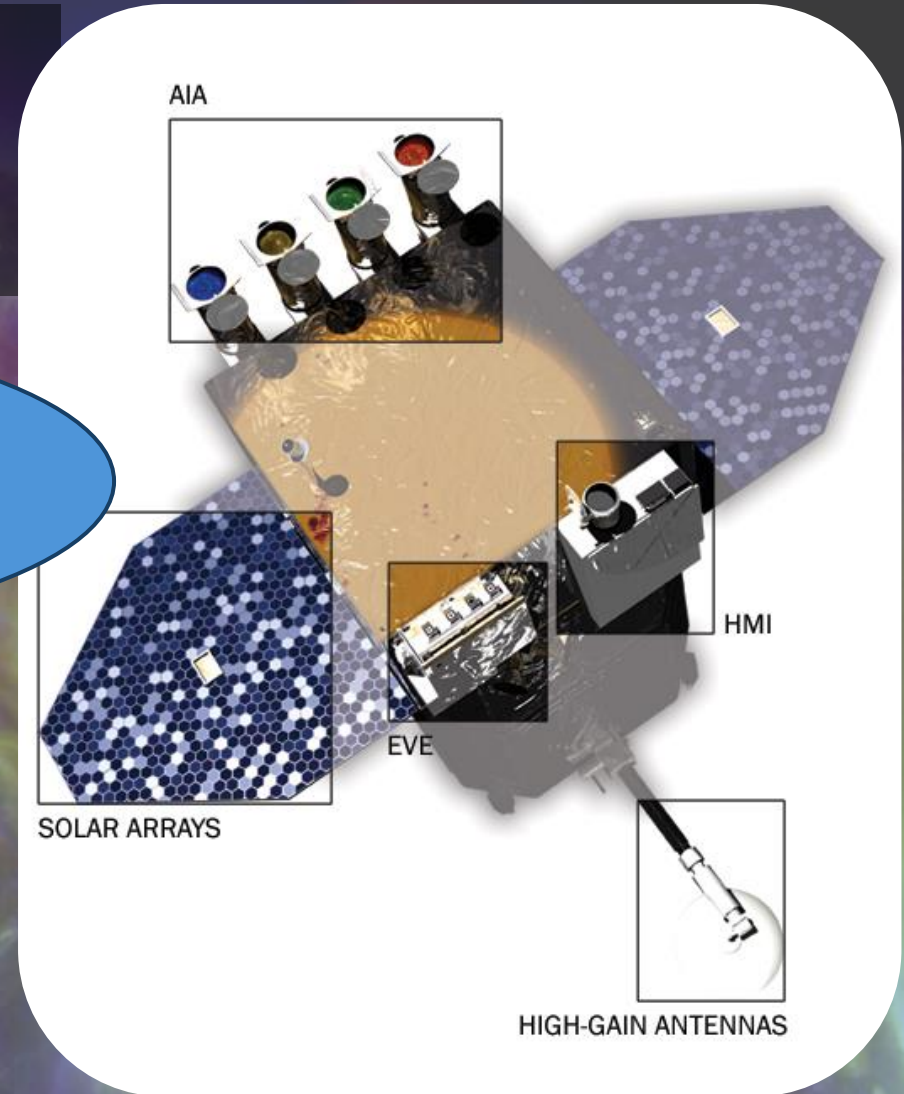
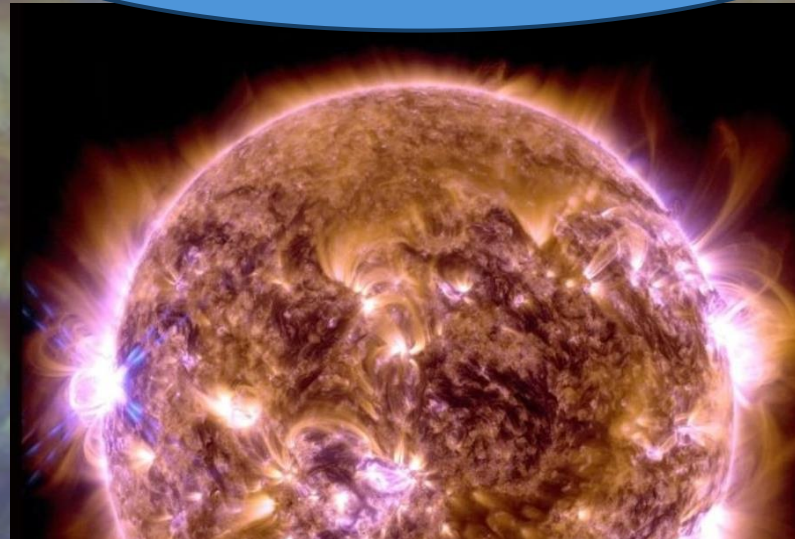
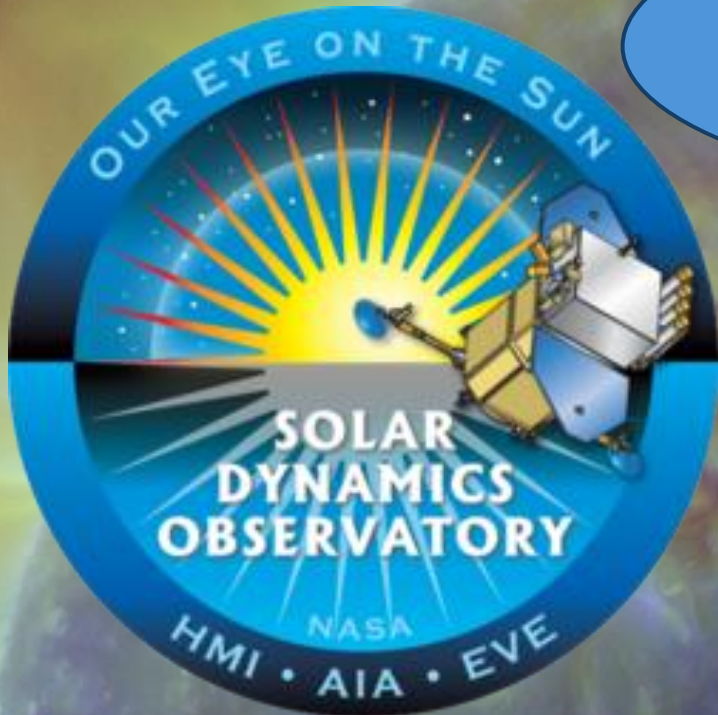
Exploring the solar surface



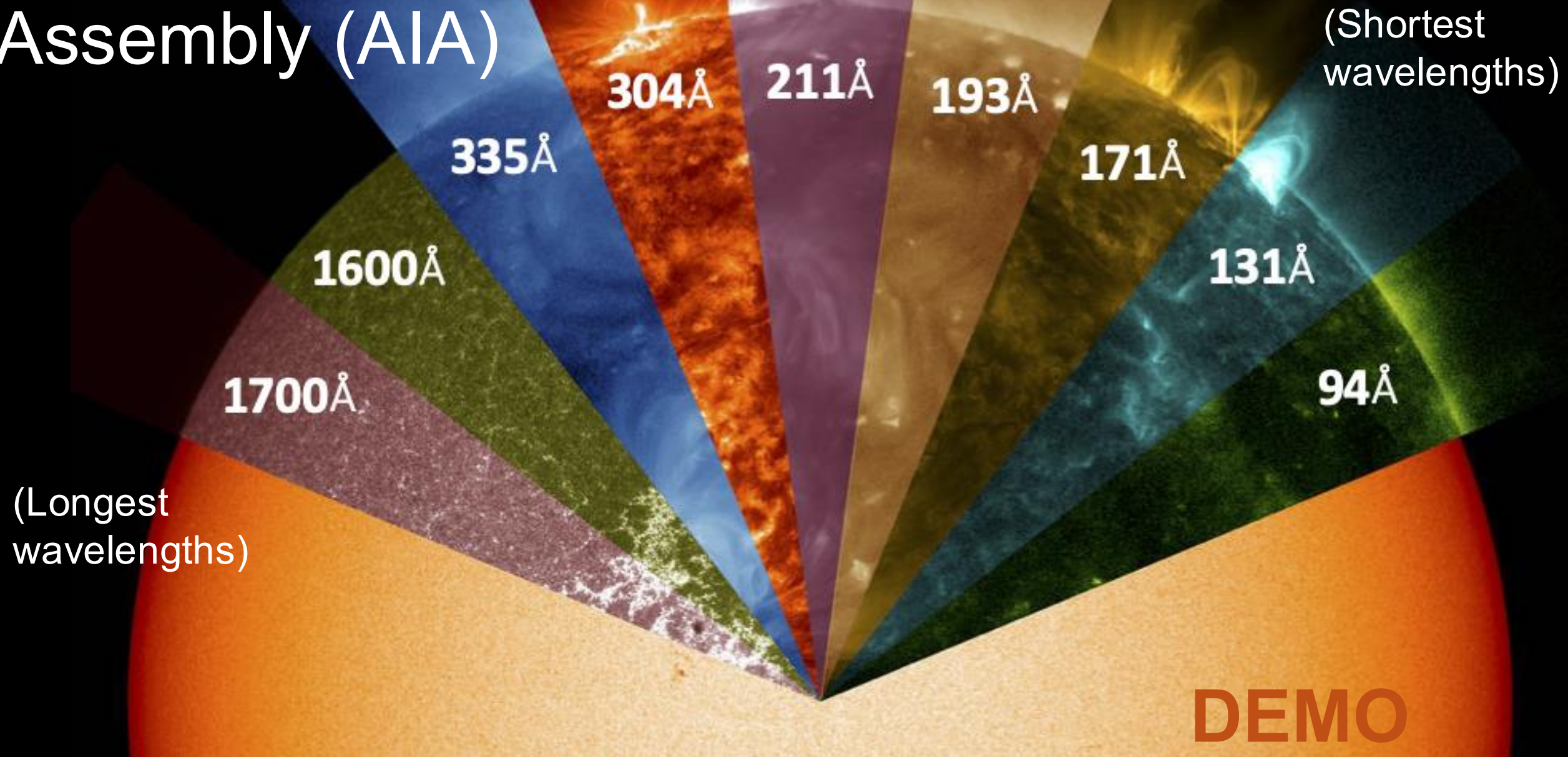
Solar Dynamics Observatory (SDO)

- Atmospheric Imaging Assembly (AIA)
- Helioseismic and Magnetic Imager (HMI)
- EUV Variability Experiment (EVE)

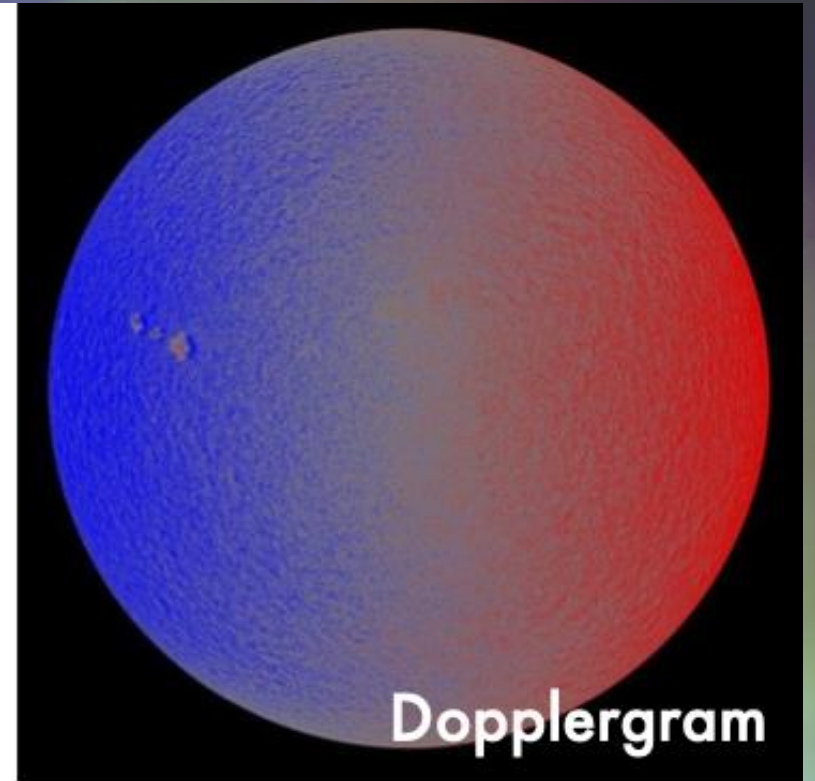
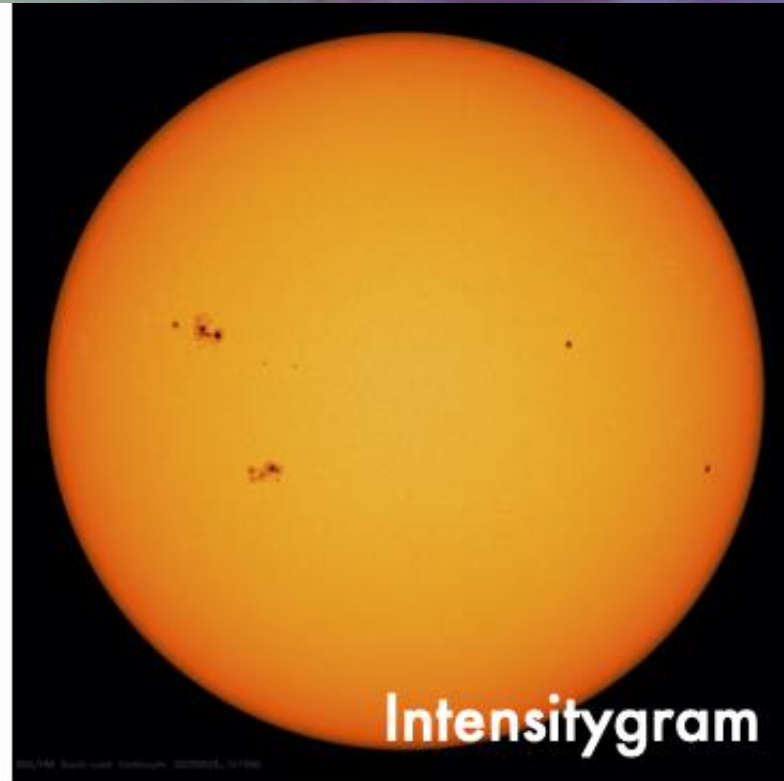
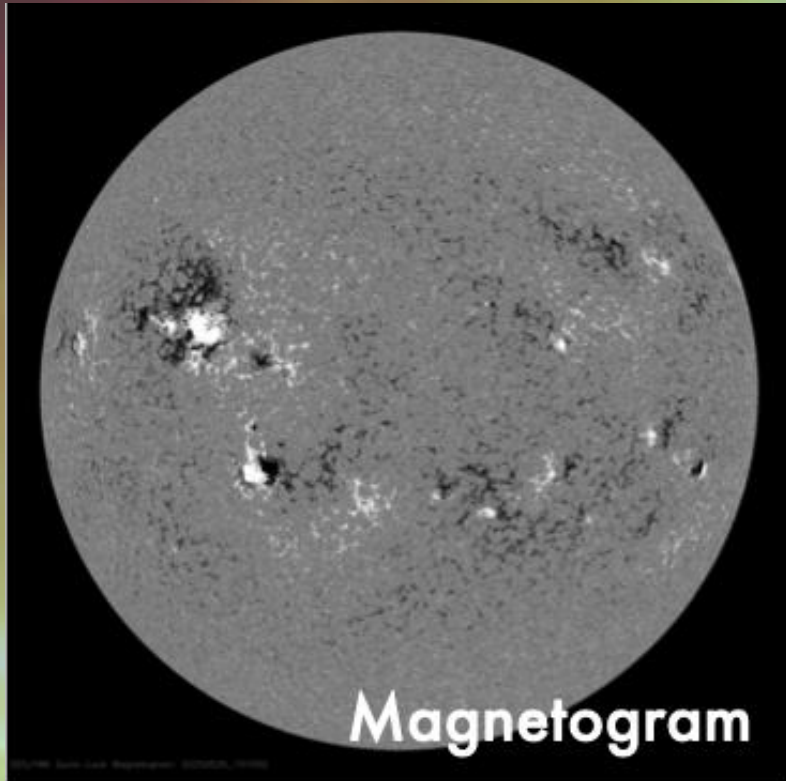
Launched in 2010, it is still going strong today!



Atmospheric Imaging Assembly (AIA)



Helioseismic and Magnetic Imager (HMI)



▼ SOLAR II - Exploring the solar surface

Since our Sun is our nearest star we can observe its surface in great detail. In February 2010, NASA launched the **Solar Dynamics Observatory (SDO)** as part of their Living With a Star Program, designed to understand the causes of solar variability and its impacts on Earth. To achieve this, SDO has three instruments on board:

- Atmospheric Imaging Assembly (AIA)
- Helioseismic and Magnetic Imager (HMI)
- EUV Variability Experiment (EVE)

GOAL : In this session we are going to be using data from two of SDO's instruments, AIA and HMI, to explore how the surface of the Sun changes overtime.

AIM : To understand how to access, manipulate and plot images of the solar surface.

OUTPUT : Create maps and animations of regions of the solar surface which display different types of phenomena.

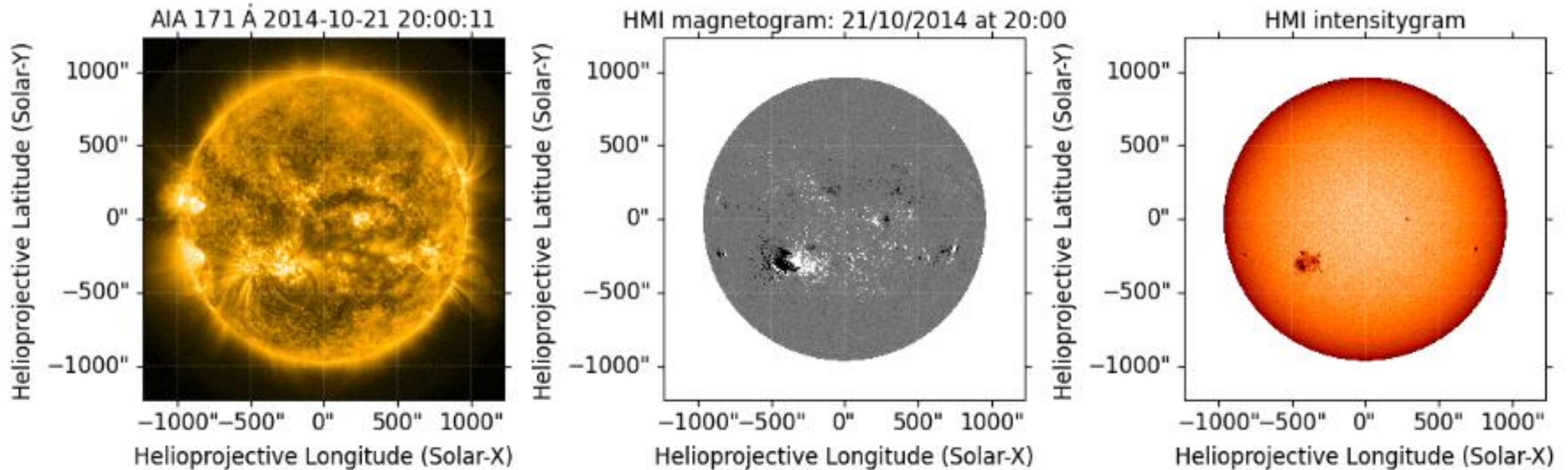
In this session we will be downloading files from the archive online, so we need to have our python session configured to know where these files are.

What is a Path?

A **path** is like an address for a file.

- **Relative path** → file near your notebook
 - Example: `images/sun.png` means there's a folder called `images` next to your notebook, and inside it a file called `sun.png`.
- **Absolute path** → full address of a file on your computer
 - Windows: `C:\\Users\\Lauren\\Documents\\sun.png`
 - macOS/Linux: `/Users/lauren/Documents/sun.png`

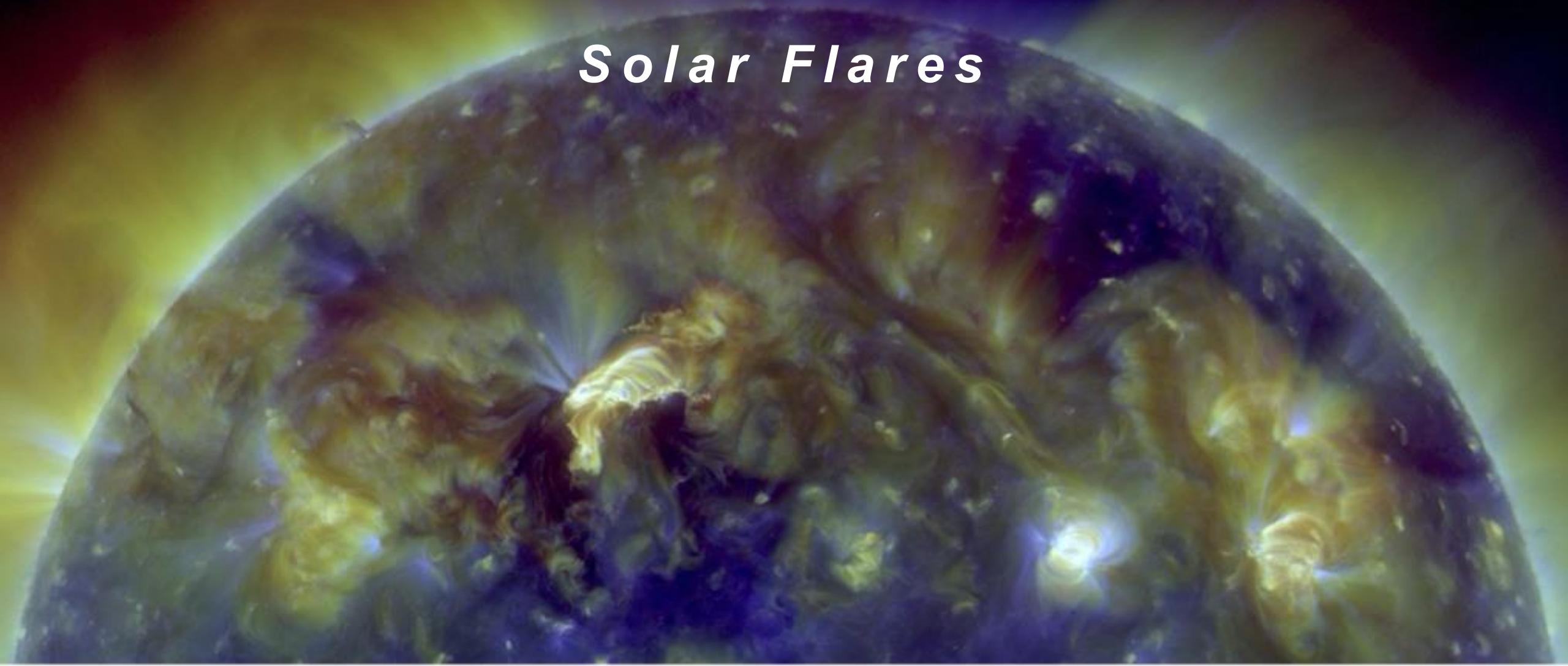
SOLAR II - Reflection



What are we seeing in each of these images and how do they relate to each other?

SOLAR III

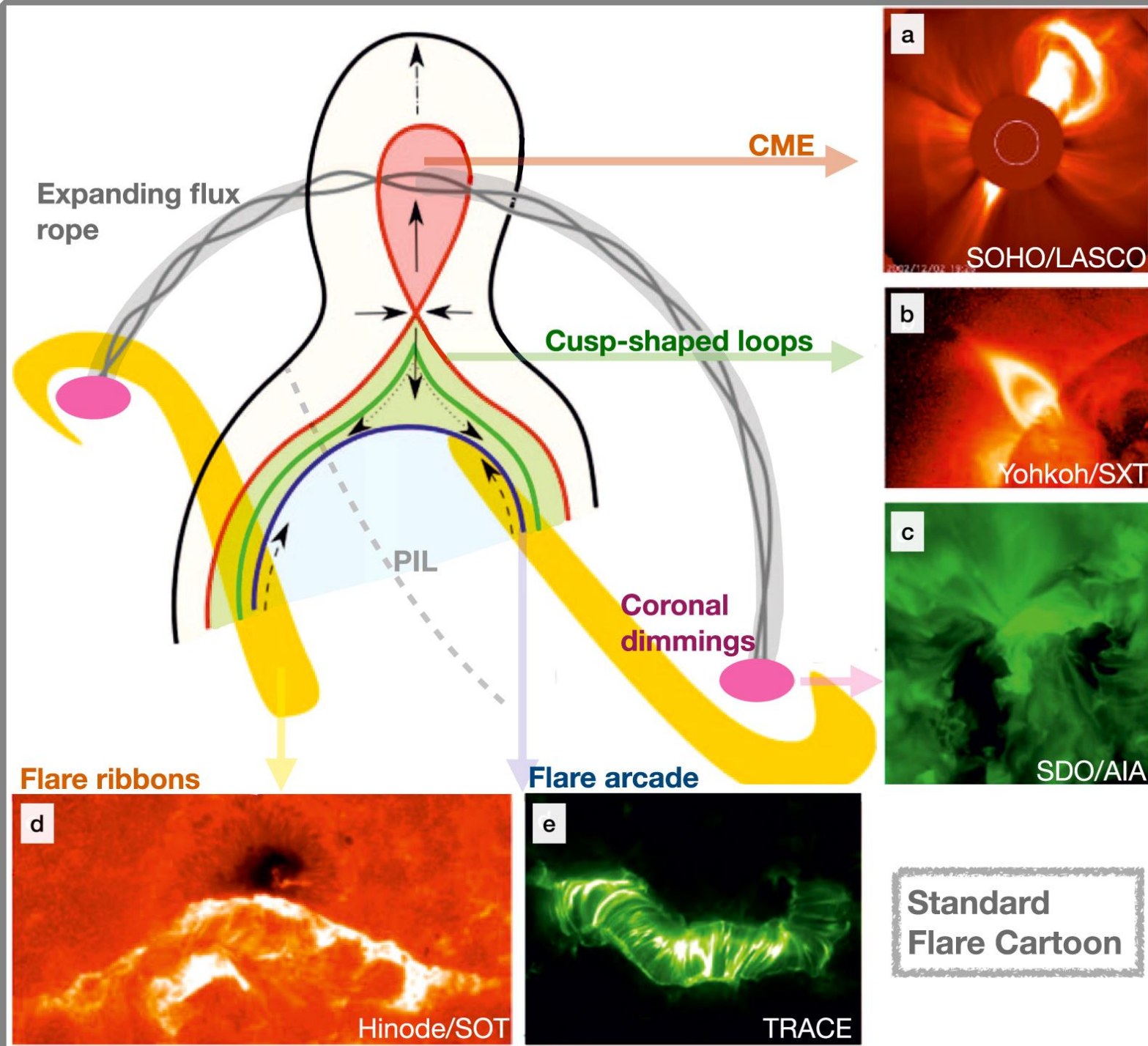
Solar Flares



Solar Flares

A solar flare is an intense, localised brightening caused by the emission of electromagnetic radiation in the Sun's atmosphere.

- Spans the entire electromagnetic spectrum
- Often accompanied by a CME.



Geostationary Operational Environmental Satellites (GOES)

- Operational since 1975
- Providing continuous imagery and data on atmospheric conditions on Earth and space weather
- collaborative National Oceanic and Atmospheric Administration (NOAA) and NASA program



GOES uses its onboard X-Ray Sensors (XRS) to take measurements of solar X-rays in the 1-8 Å (long or xrsb channel) and 0.5-4.0 Å (short or xrsa channel) passbands

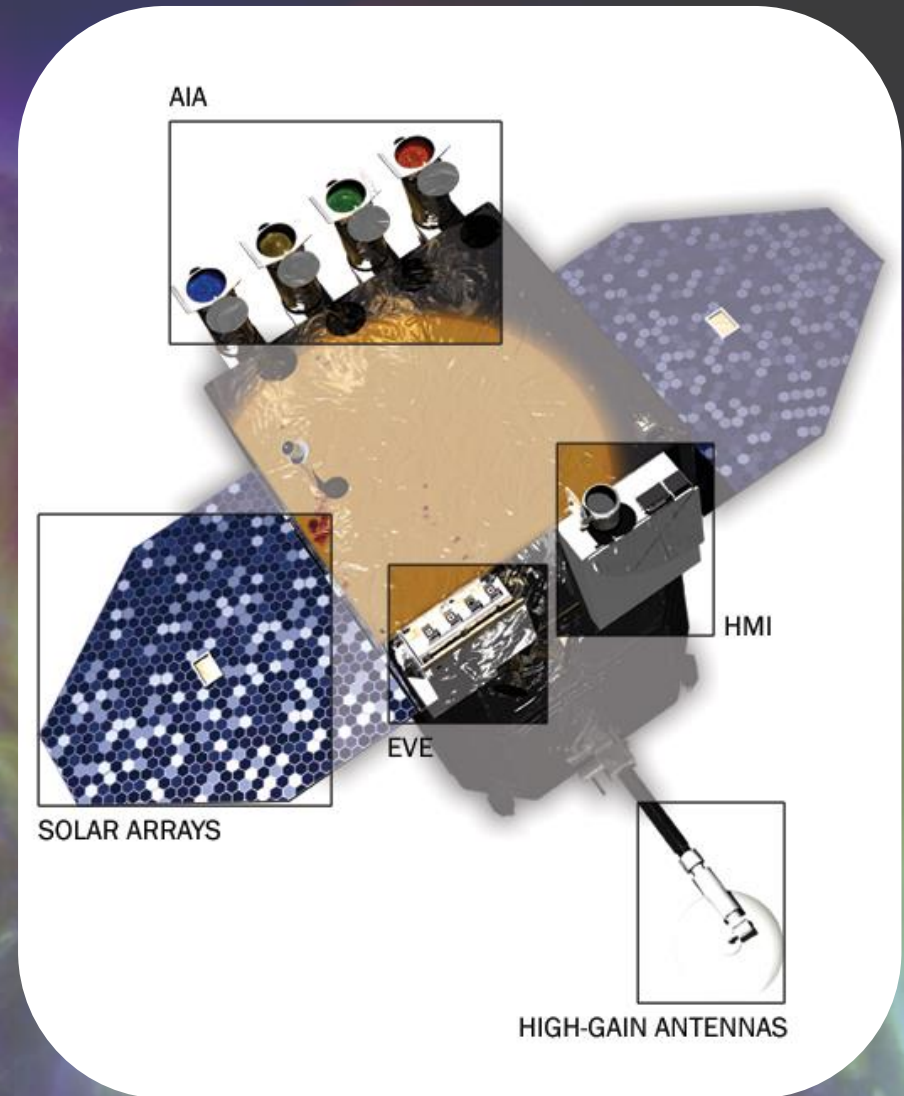


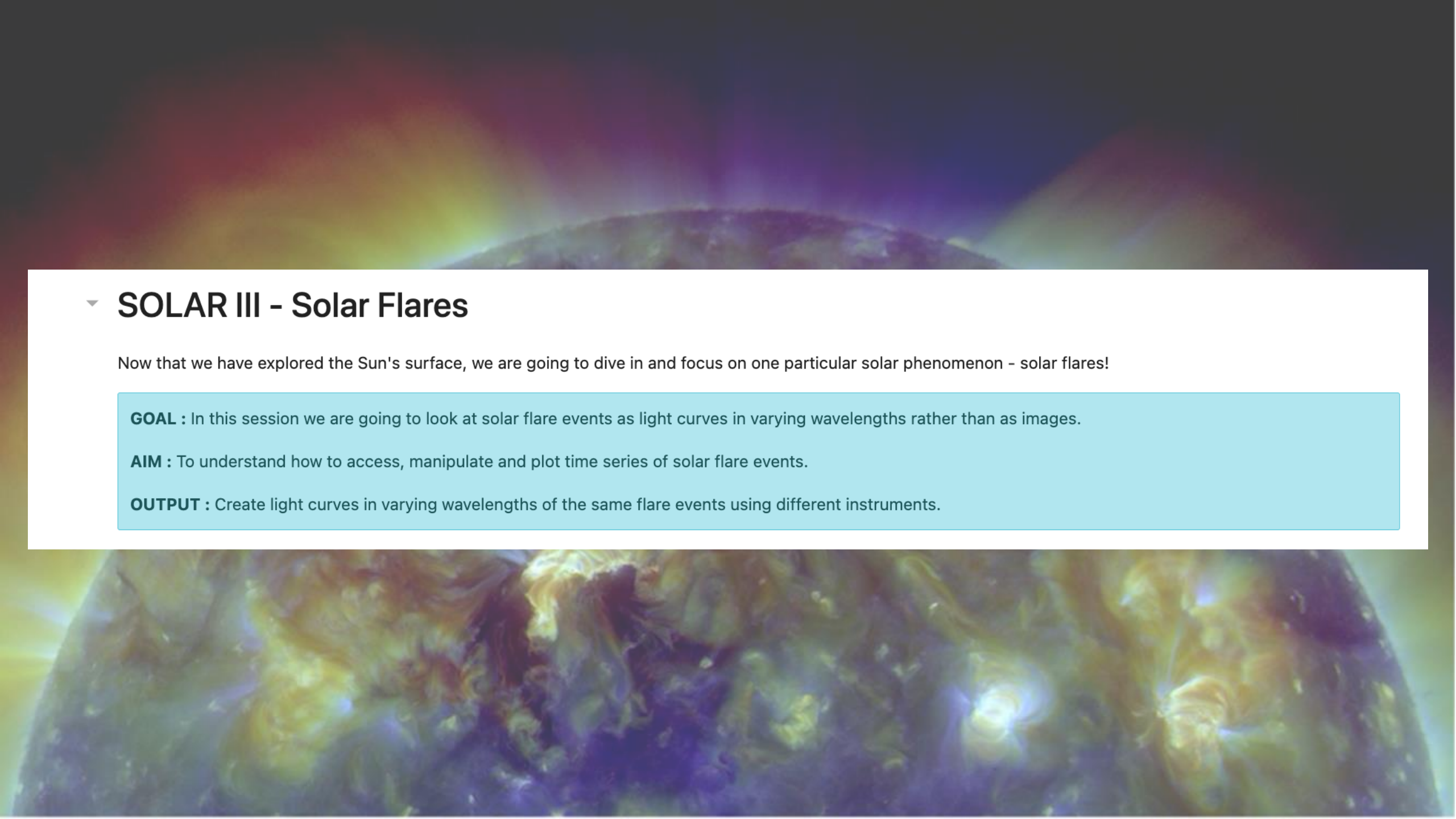
Extreme ultraviolet Variability Experiment (EVE)

EVE records the Sun's extreme ultraviolet (EUV) radiation — light with very short wavelengths (roughly 0.1 to 105 nm) that we cannot see with our eyes.



- It takes measurements every 10 seconds
- It provides a full EUV spectrum
- It is continuous, high-cadence data.





▼ SOLAR III - Solar Flares

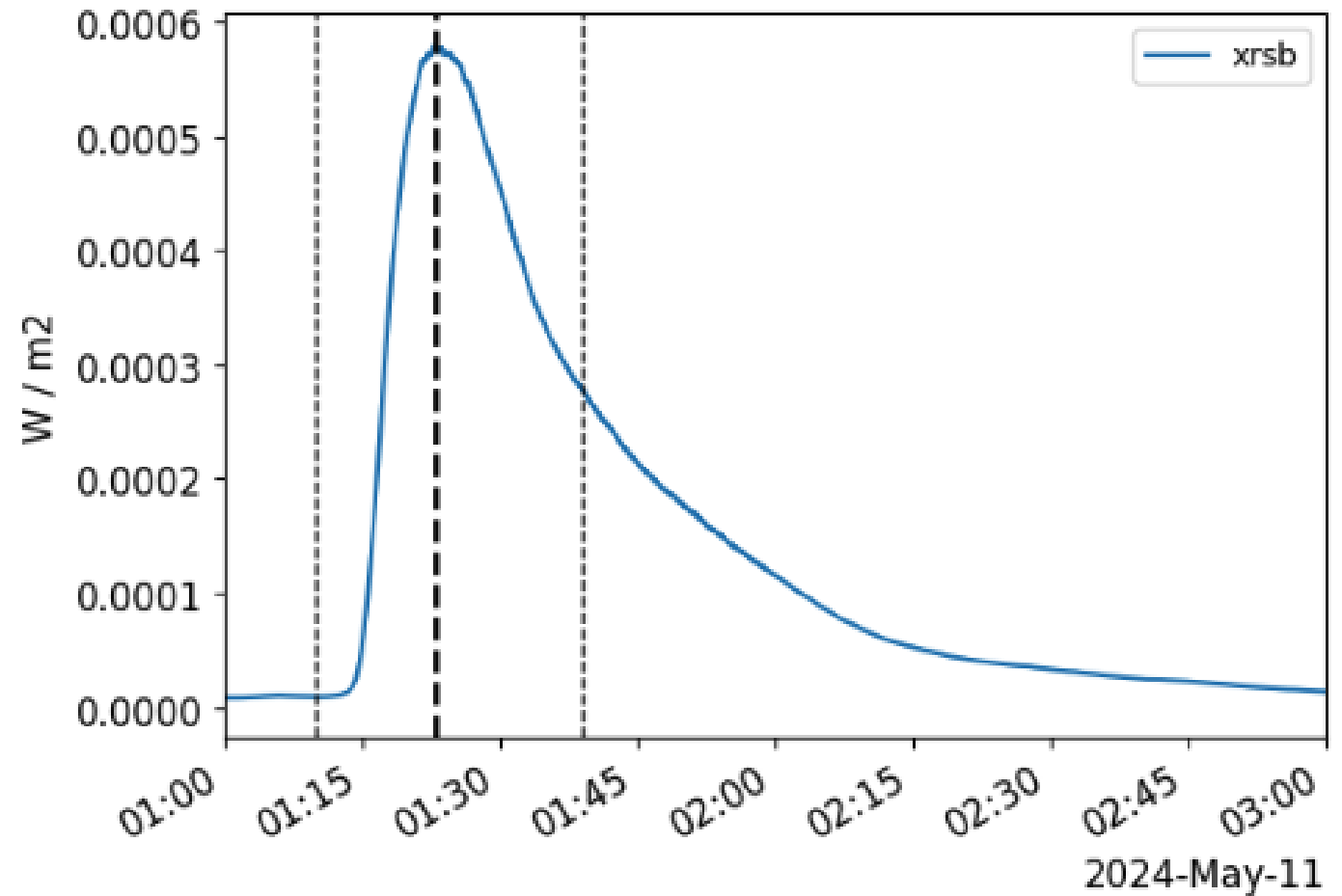
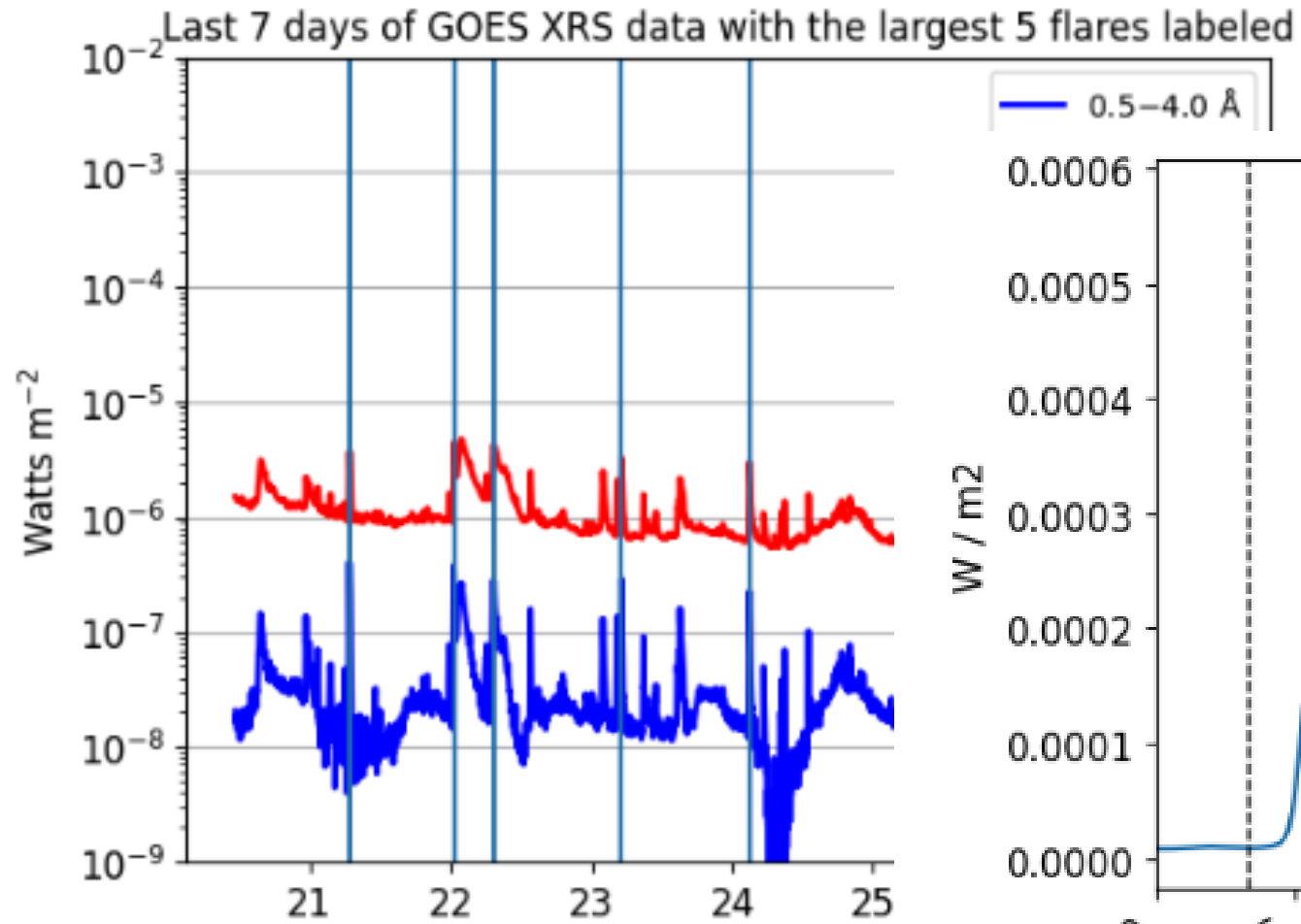
Now that we have explored the Sun's surface, we are going to dive in and focus on one particular solar phenomenon - solar flares!

GOAL : In this session we are going to look at solar flare events as light curves in varying wavelengths rather than as images.

AIM : To understand how to access, manipulate and plot time series of solar flare events.

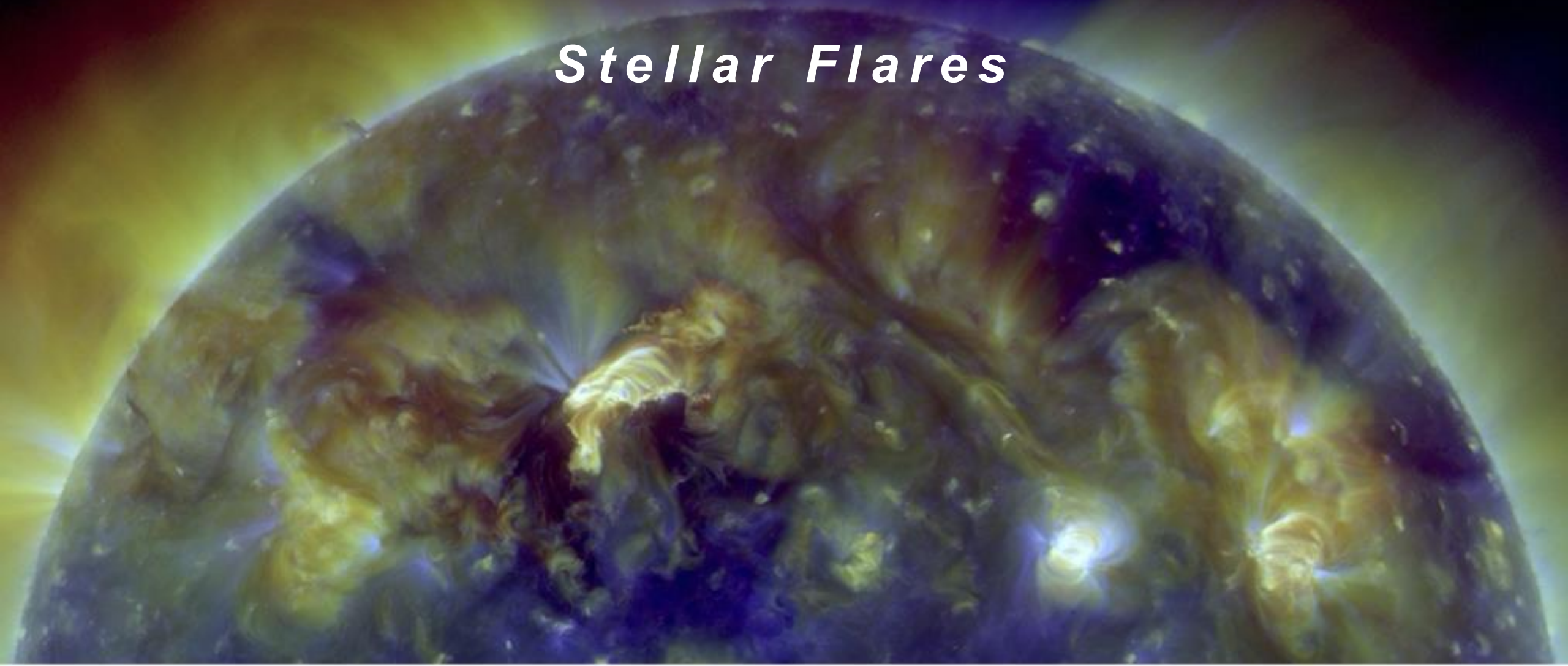
OUTPUT : Create light curves in varying wavelengths of the same flare events using different instruments.

SOLAR III - Reflection

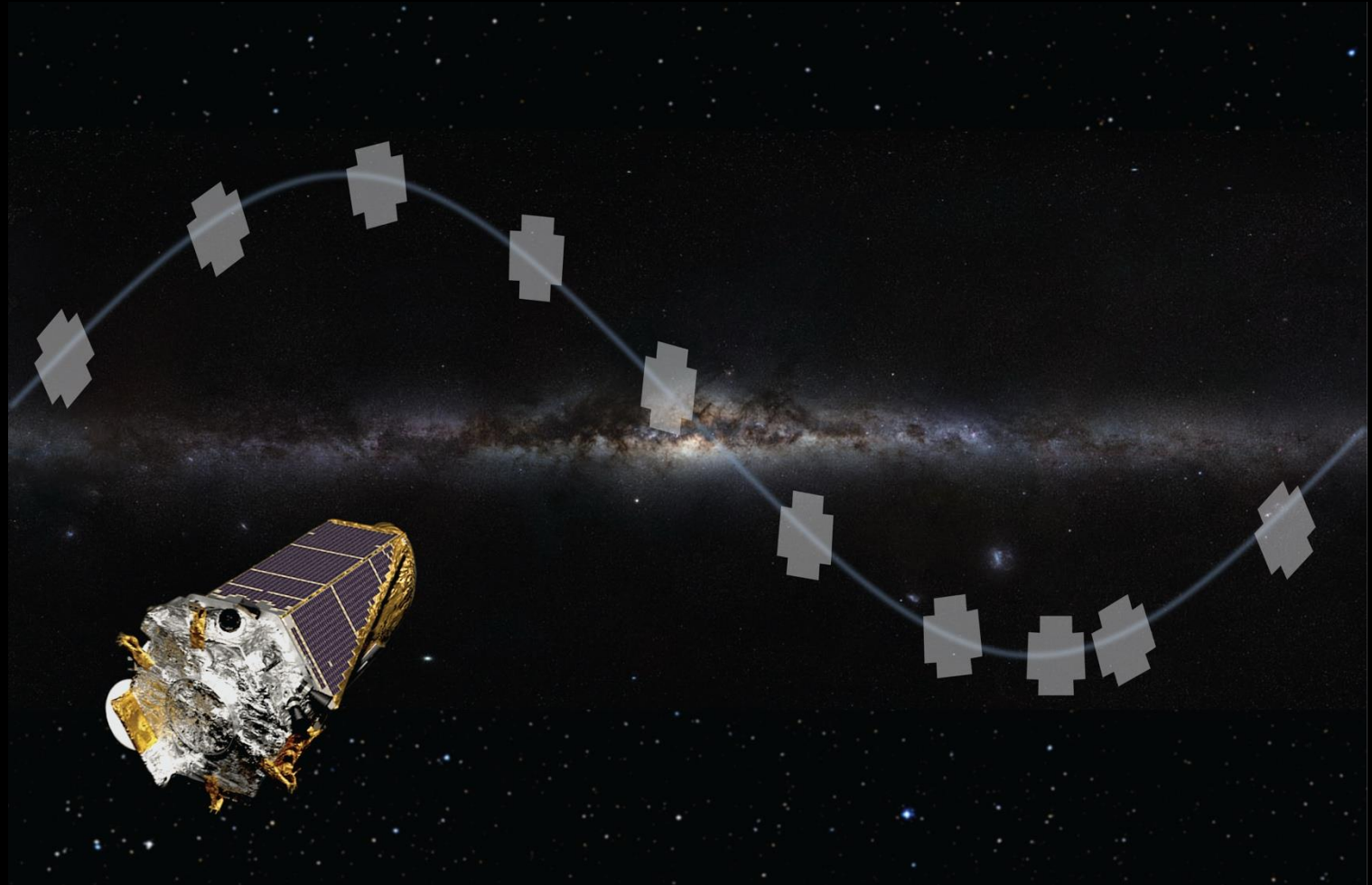
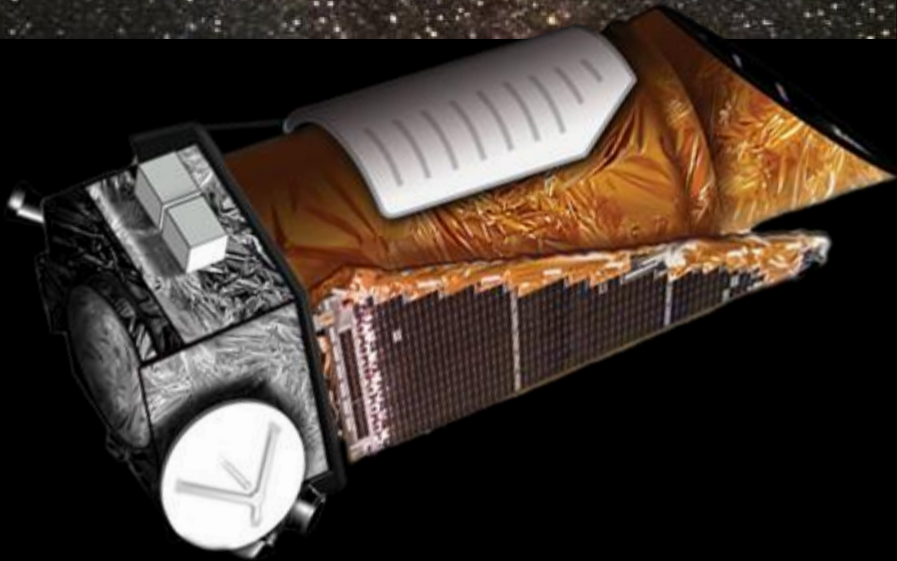
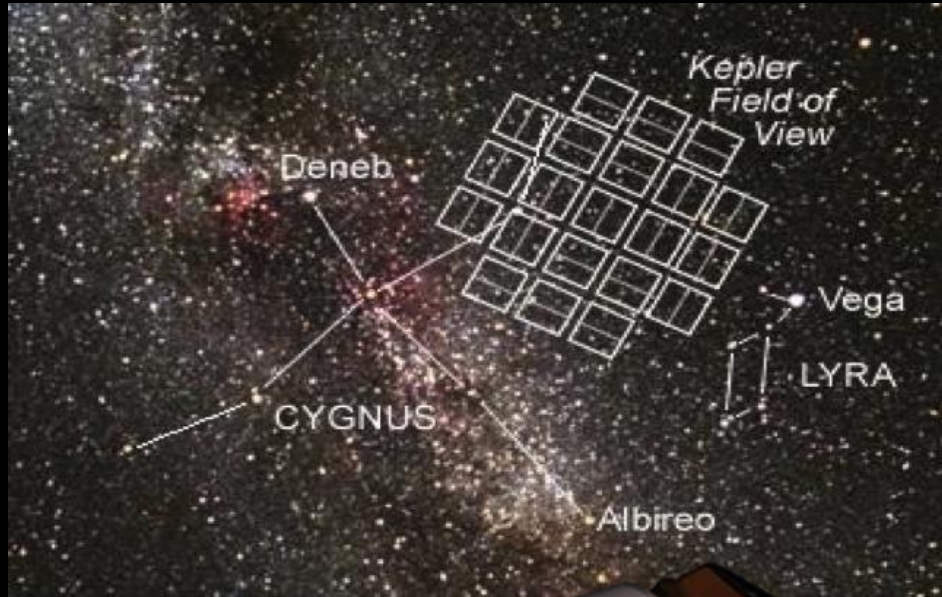


SOLAR IV

Stellar Flares

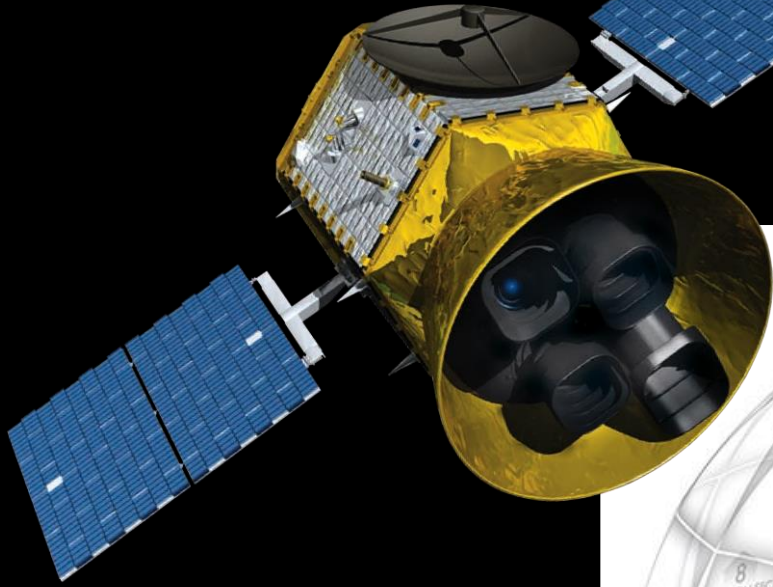


Kepler/K2

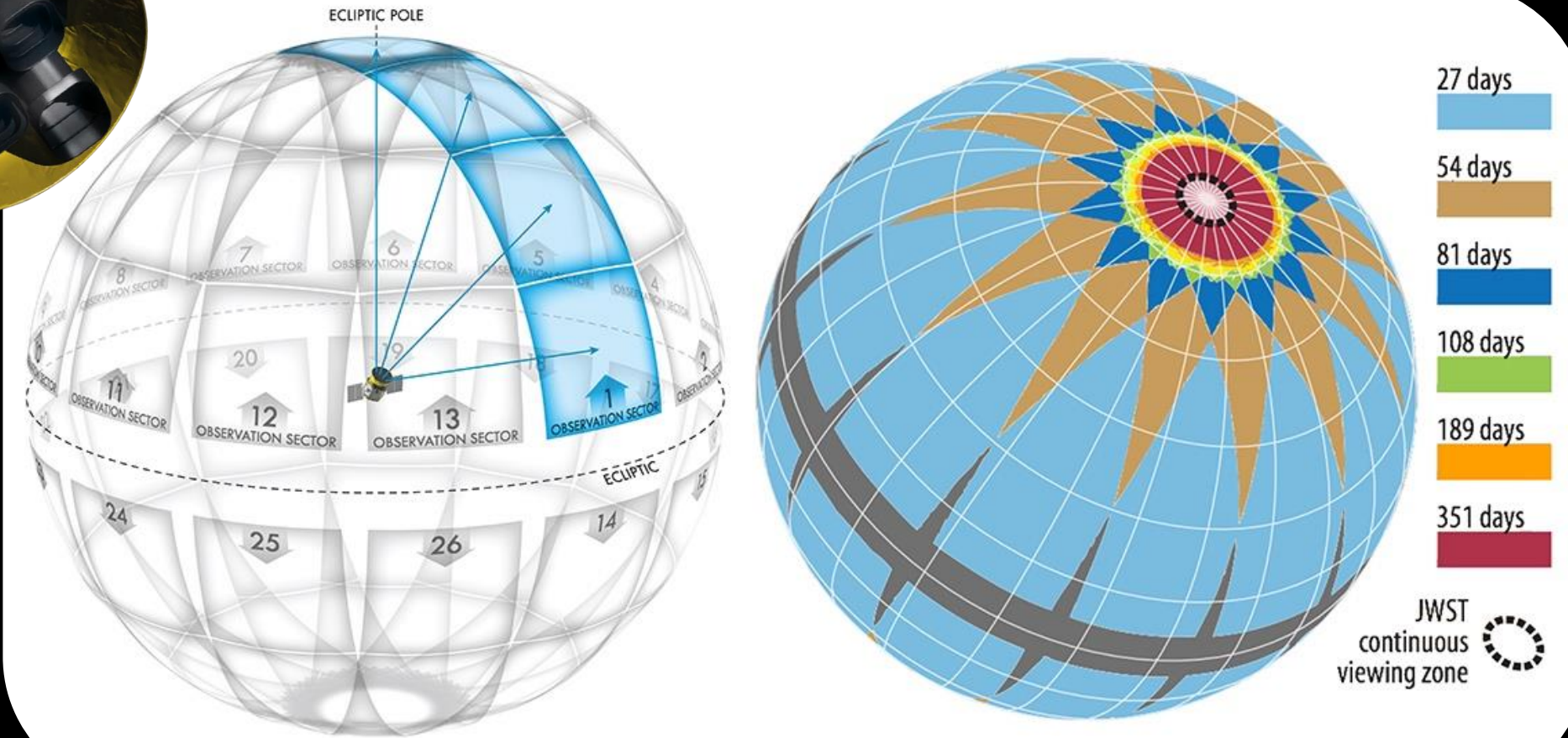


Kepler allowed for uninterrupted observations of hundreds of thousands of stars of many types.

Transiting Exoplanet Survey Satellite (TESS)



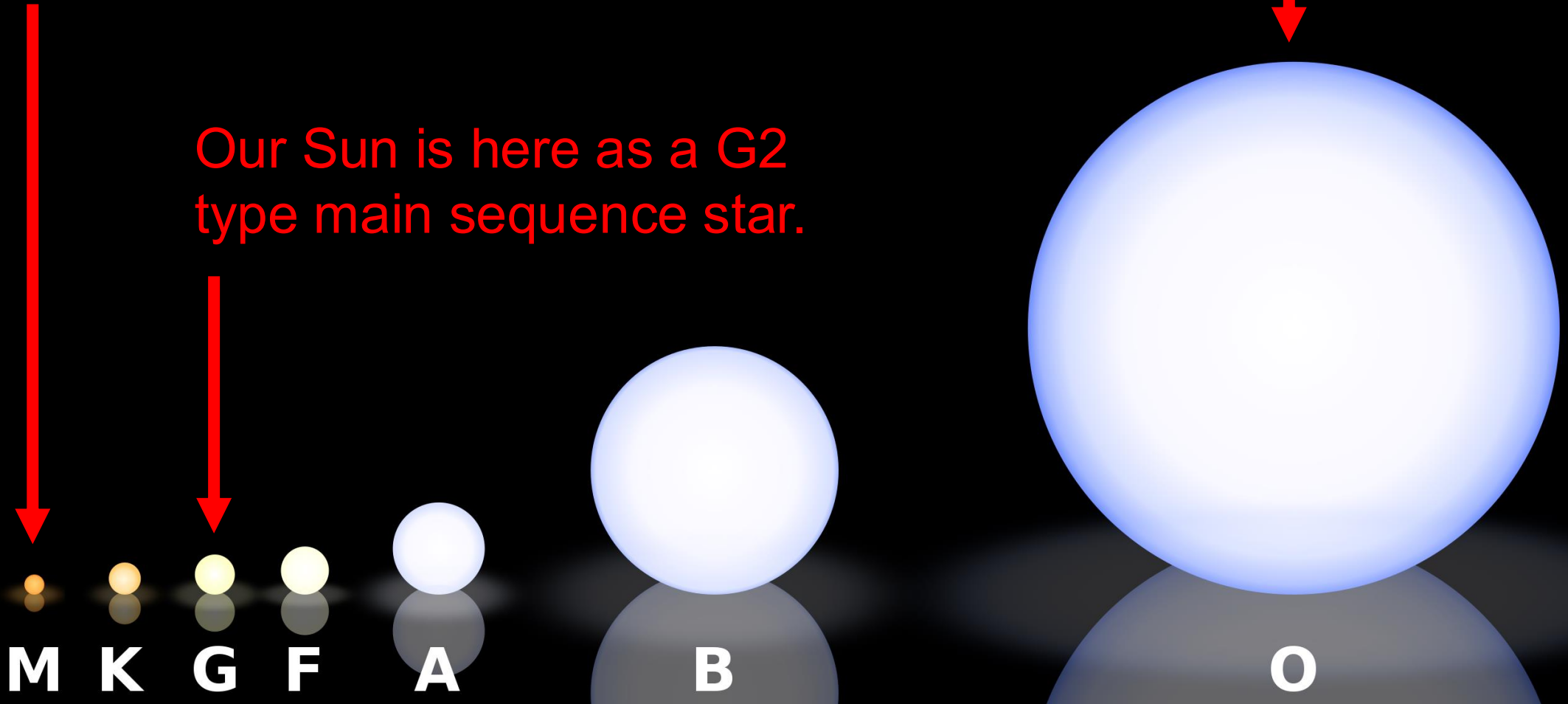
Known as the Planet Hunter, TESS observes in sectors for a duration of 27 days.

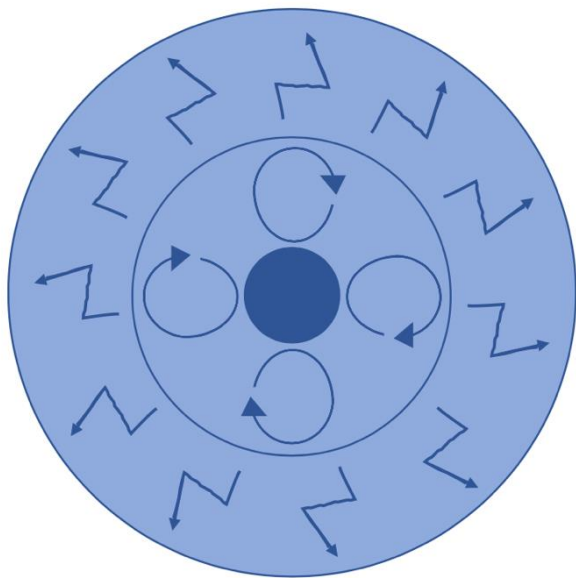


M dwarfs here are the most abundant star in the solar neighbourhood.

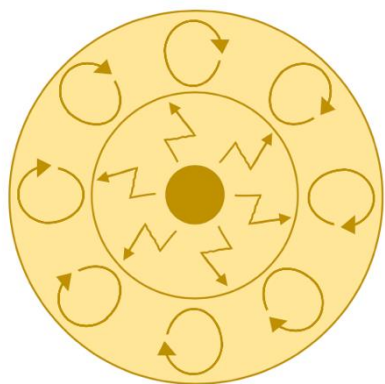
Our Sun is here as a G2 type main sequence star.

The most massive and hottest stars in the universe.
Live Fast, Die Young!

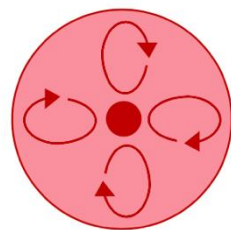
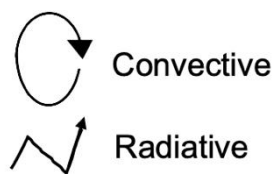




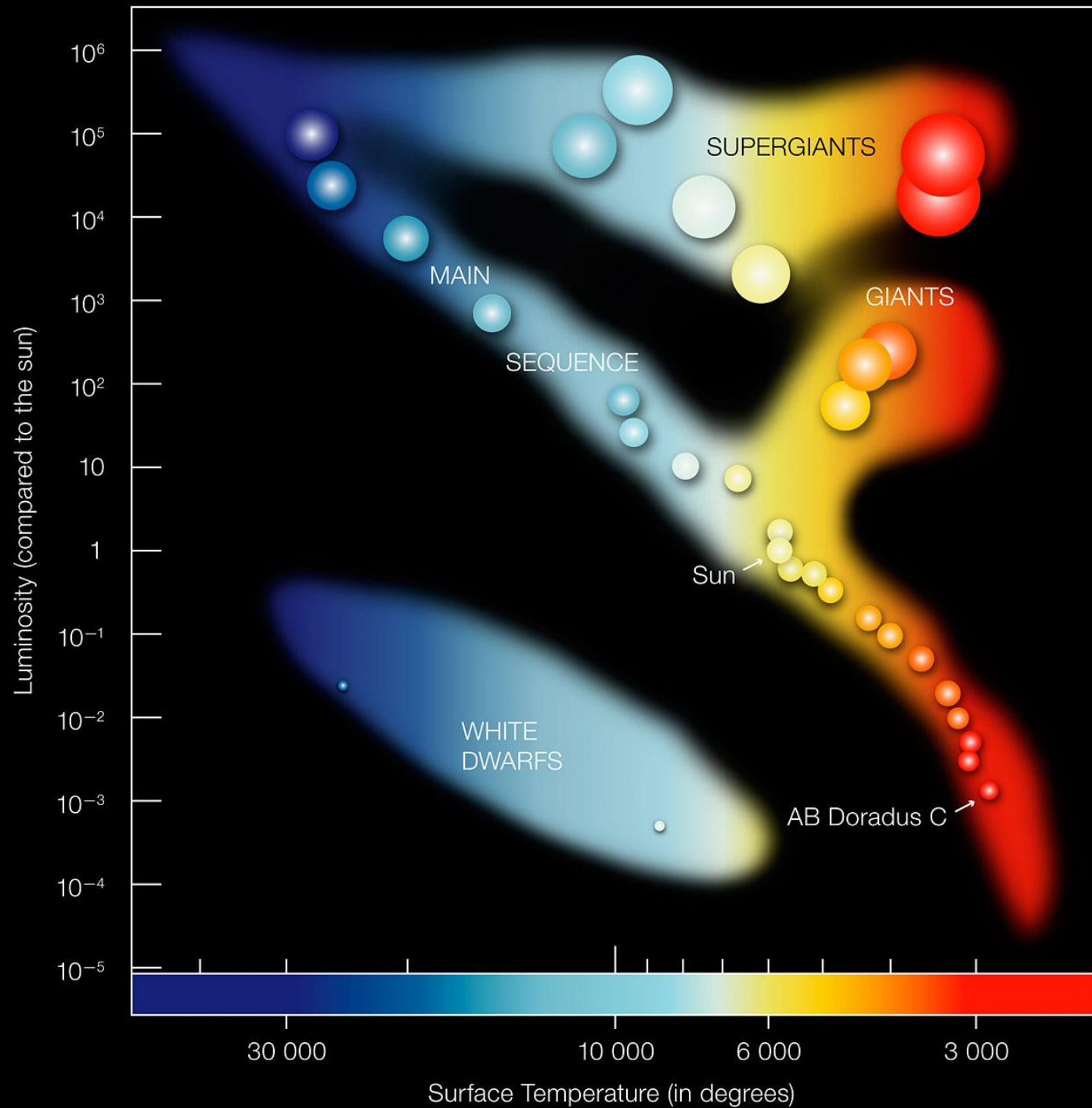
$M_* > 1.5 M_\odot$



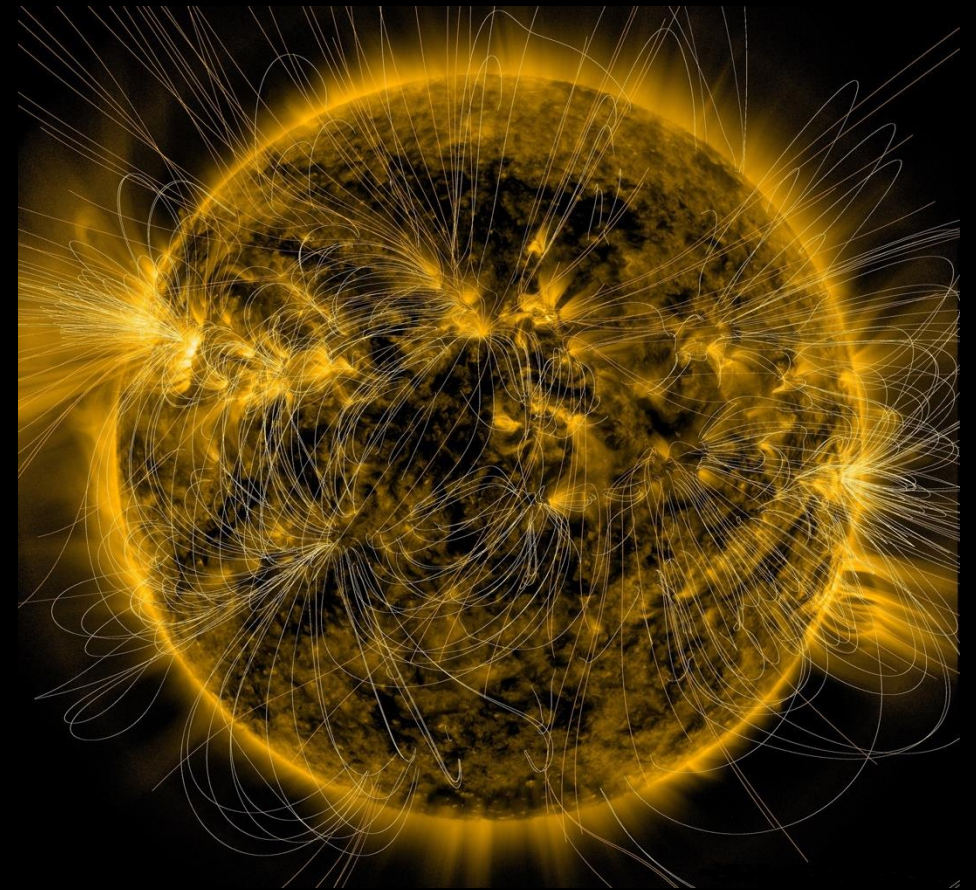
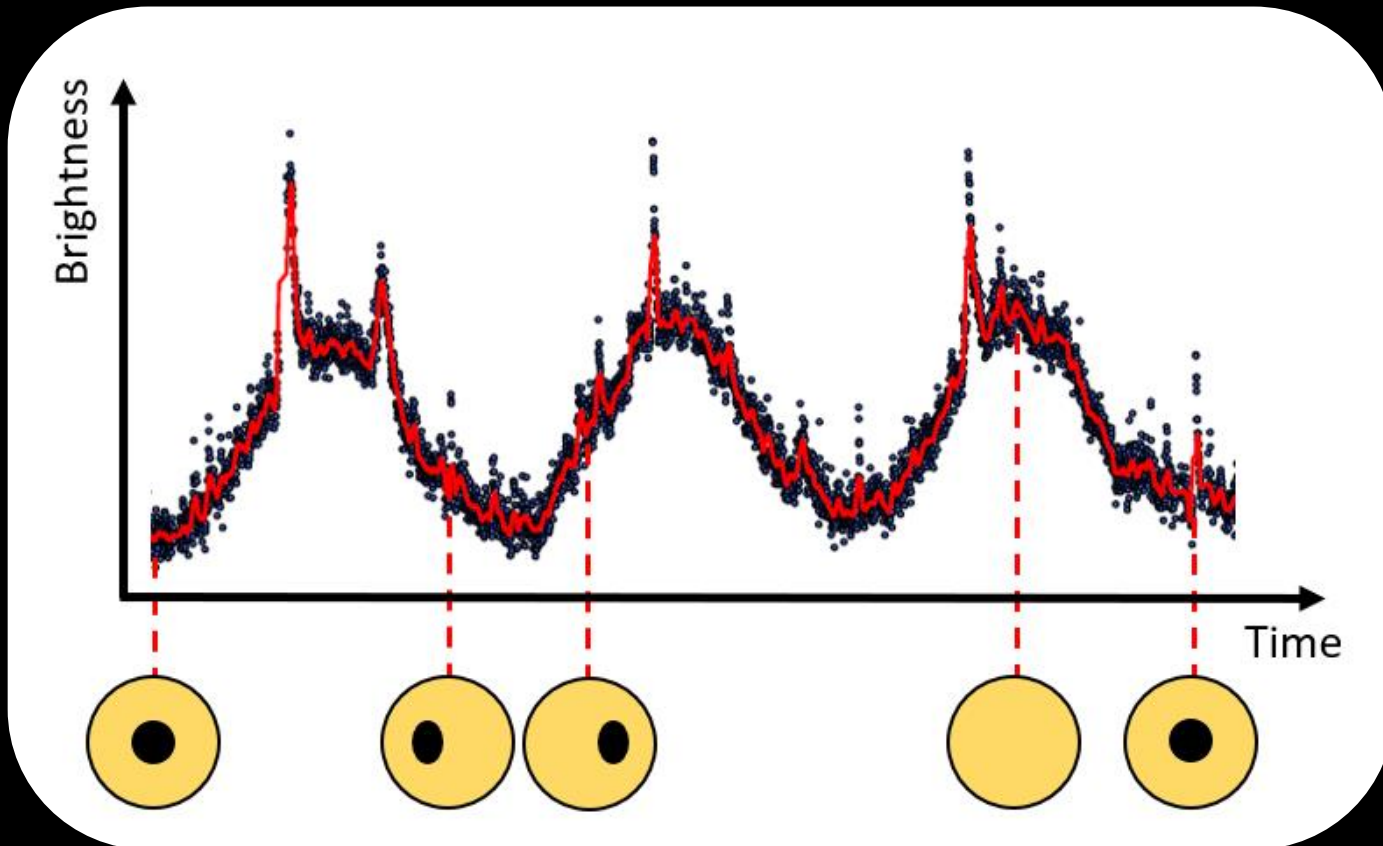
$0.3 M_\odot < M_* < 1.5 M_\odot$



$M_* < 0.3 M_\odot$

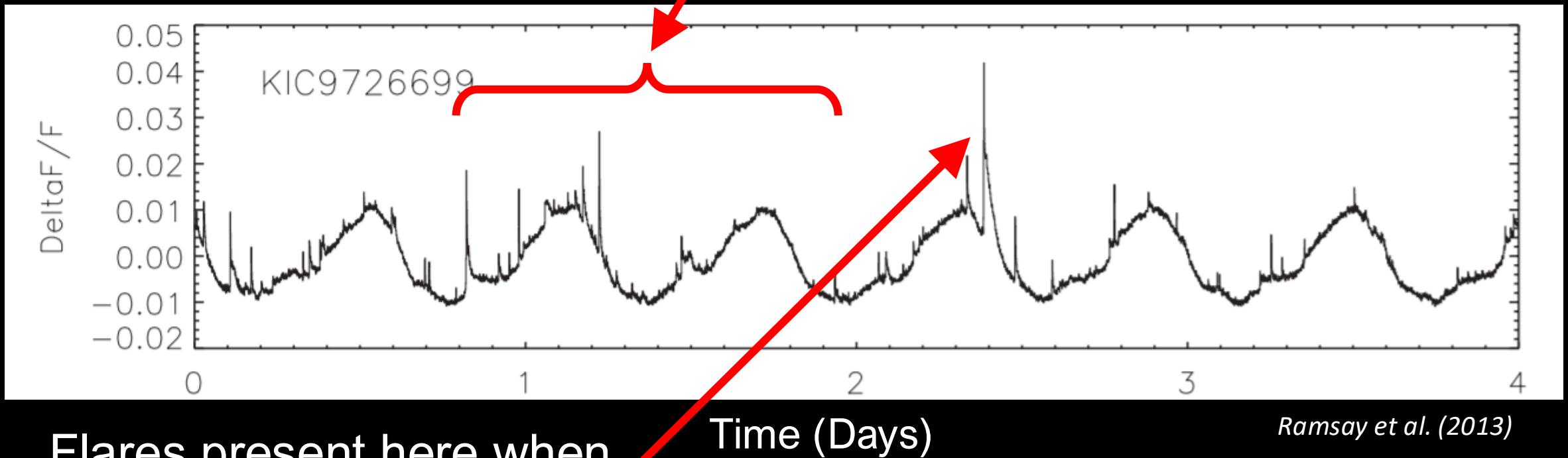


There are many other phenomena which can contribute to a light curve making it more difficult to pick out an exoplanet transit signal.



What does this look like?

Clear rotation modulation caused by the presence of a long-lived star spot.



Flares present here when the star spot is least visible.

Ramsay et al. (2013)

▼ SOLAR IV - Stellar Flares

So far, we have been investigating solar activity with a focus on solar flares. However, we also know that other stars can also show similar behaviour, and this has become a rapidly growing field within astronomy. Therefore, in this session, we will focus on looking at stellar flares on stars outside our solar system!

GOAL : In this session, we are going to be using data from TESS, to investigate stellar flares on other stars.

AIM : To understand how to access, manipulate and plot timeseries data of other stars from space-based missions.

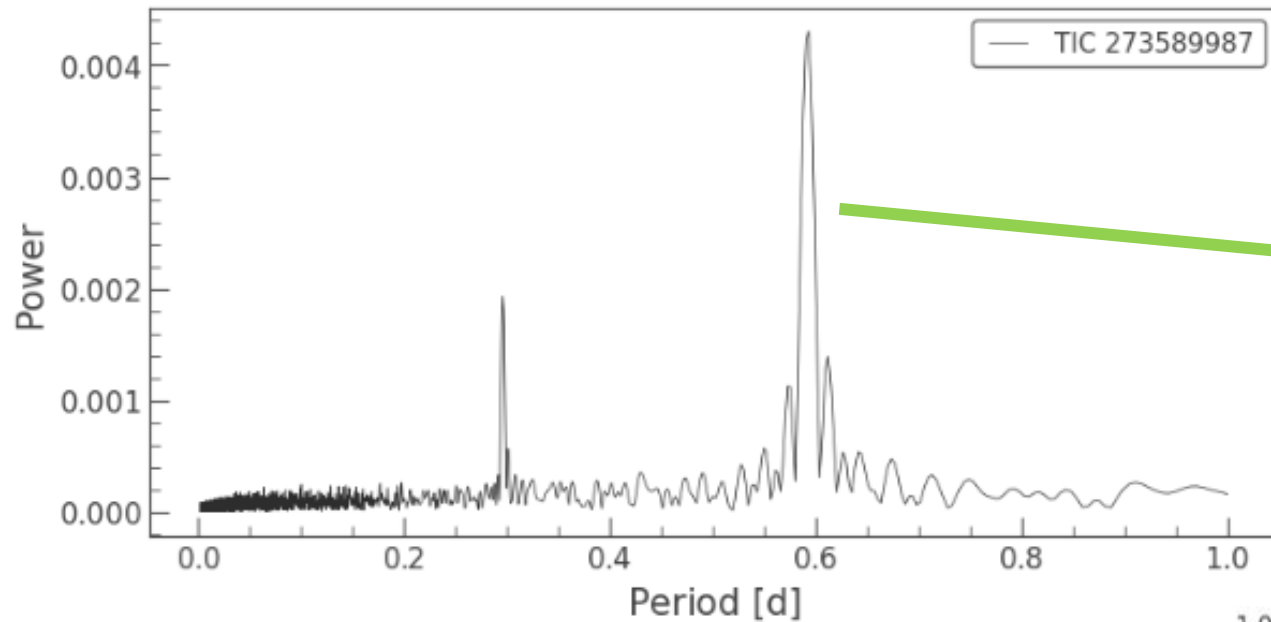
OUTPUT : Create light curve plots, determine the rotation period and activity levels for a sample of stars.

To achieve our goals and aims, we will be using the Python package [Lightcurve](#), which is a powerful tool for downloading, manipulating and analysing space-based data from missions such as Kepler and the Transiting Exoplanet Survey Satellite (TESS).

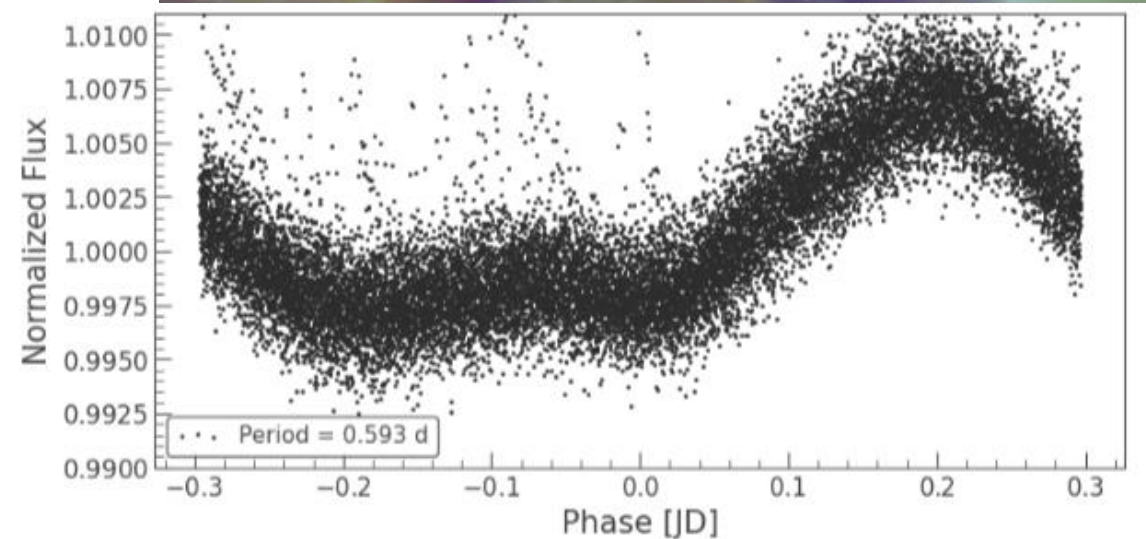
If you haven't already, you can install the `lightcurve` Python package by running the bash command:

```
pip install lightcurve
```

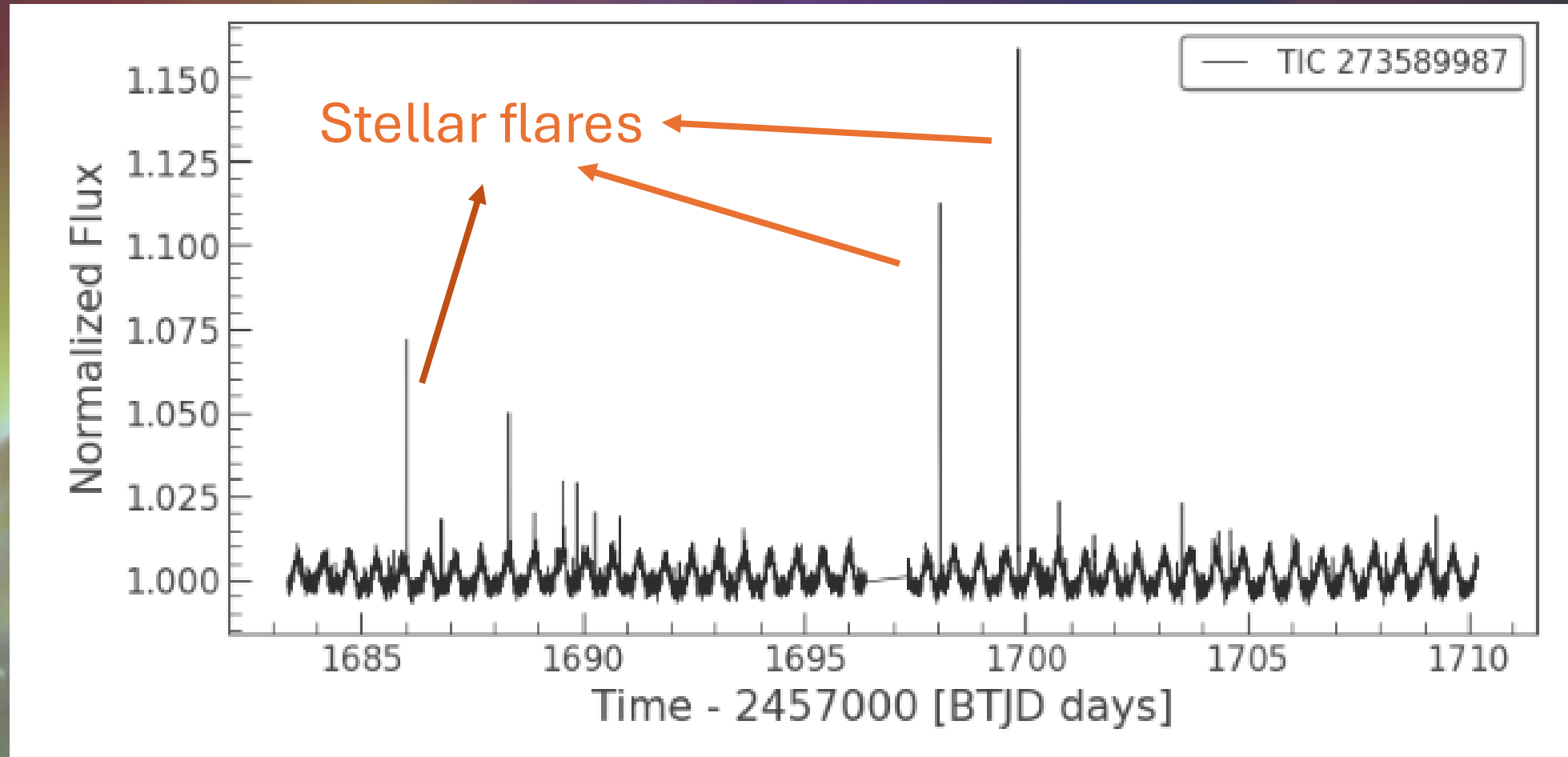
SOLAR IV - Reflection



When adding more data
did the rotation period
change?



SOLAR IV - Reflection



What classification did these flares have when compared to solar?